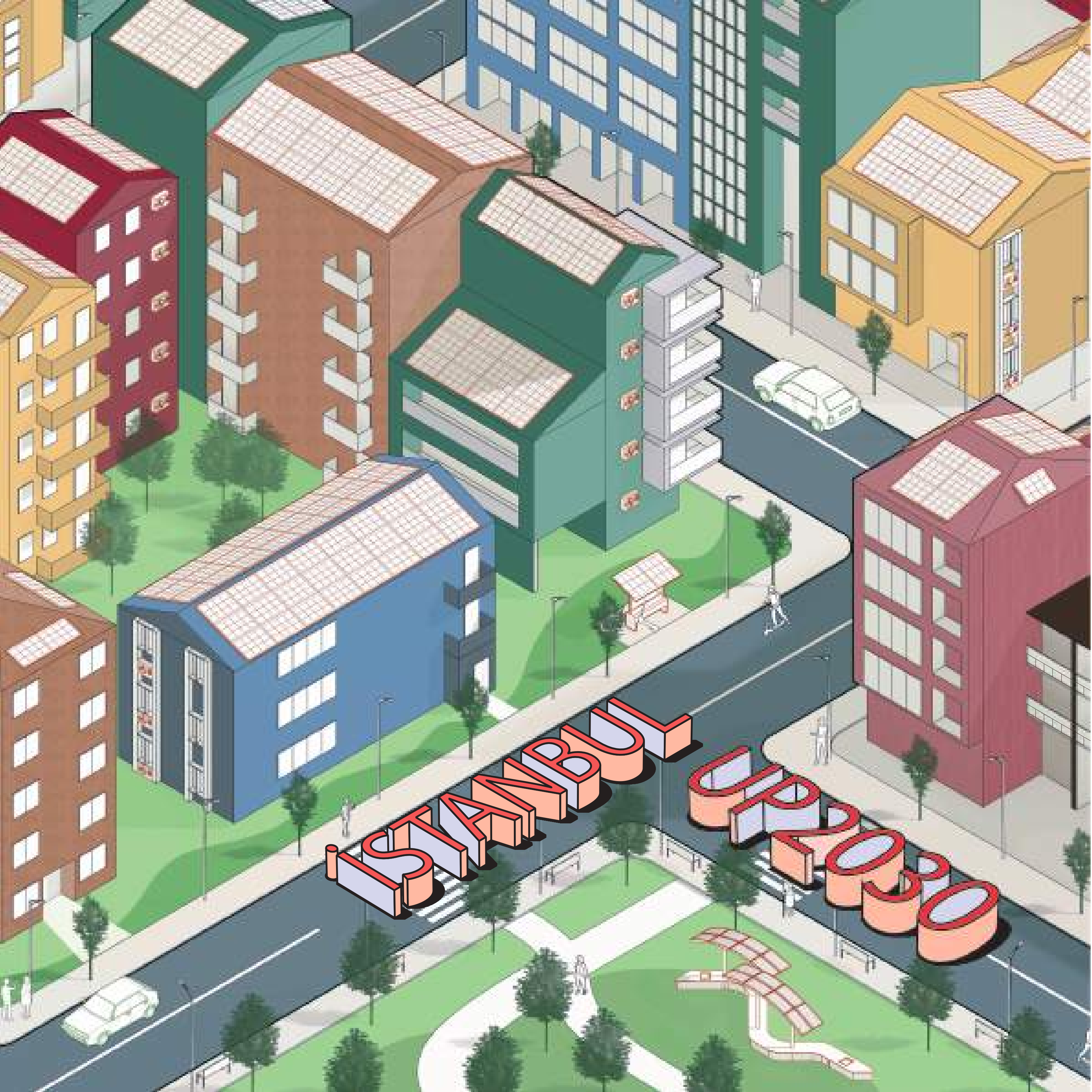


İSTANBUL
UP2030

Content

1. UP2030 PROJECT

1.1.General View

UP2030 aims to help cities achieve their climate neutrality goals by using urban planning and design as key tools for transformation. The project assists city stakeholders and local governments in integrating climate neutrality into everyday practices and long-term strategies. To support the clean energy transition, an innovative methodology, known as the **5UP approach**, is developed and implemented, combining scientific insights with practical tools and methods through collaborative development.

UPDATE	planning and design approaches, standards, codes and policies for urban transformations
UPSKILL	city's stakeholder ecosystem to co-develop urban planning and design enabled transformation pathway
UPGRADE	our neighborhoods using built & natural environment prototypes, supportive models & tools for planning and design
UPSCALE	governance arrangements, financial mechanisms, policy development & decision-making for urban planning
UPTAKE	activities to raise awareness and transfer knowledge across European cities and beyond

UP2030 is concerned with mainstreaming the climate neutrality agenda through urban planning and design to improve the overall quality of life in cities. The project strategically focuses its prototypes on the **neighborhood level**, as this level is crucial for addressing challenges, promoting reinvestment, and fostering climate innovation. UP2030 will support local governments in creating an innovation-enabling city environment through the development of appropriate policy frameworks, inclusive participatory processes, promotion of sustainable behaviours, capacity building within municipal departments, new governance models, and financial support mechanisms. Ultimately, the project supports cities and stakeholders to align their efforts with the core values of **equity, resilience, climate neutrality, and sustainability.**

UP2030 is implemented in several pilot cities across Europe and beyond, with each city adapting the solutions to its local context. These cities include **Belfast, Budapest, Granollers, Istanbul, Lisbon, Milan, Muenster, Rio de Janeiro, Rotterdam, Thessaloniki, and Zagreb.** By piloting innovative urban planning and sustainability strategies, these cities serve as models for future climate-neutral urban development [1].

1 *UP2030 Project. About the Project.* https://up2030-he.eu/about/

1.2. Climate Change

Climate change refers to long-term shifts in temperatures and weather patterns. These shifts can be natural, such as those caused by changes in solar activity or large volcanic eruptions. However, since the 1800s, human activities have been the main cause of climate change, primarily through the use of fossil fuels such as coal, oil, and gas. The most important greenhouse gases include carbon dioxide and methane. These are produced, for example, by the use of petroleum and natural gas for transportation and the energy consumption of buildings [2] .

Cities play a significant role in contributing to climate change, as they are responsible for more than 70% of annual global carbon emissions and over 60% of global energy consumption [3]. Buildings account for 34% of global final energy consumption and 37% of CO_2 emissions [4]. Cities are also expected to become increasingly vulnerable to extreme climate events such as heat waves, droughts, storms, and floods as a result of climate change.

To mitigate the effects of climate change, it is necessary to reduce the energy consumption and associated CO_2 emissions of buildings, especially in highly urbanized areas. This makes the analysis of building energy performance at the urban scale crucial for stakeholders to anticipate and prepare for climate change impacts, identify vulnerabilities and risks, and evaluate potential opportunities for energy renovation strategies [5].

As the energy use of buildings is one of the main causes of climate change, energy renovation becomes essential. Energy-efficient renovation strategies are also crucial to increasing building resilience to the effects of climate change.

2 United Nations. (2025). *What is climate change?* United Nations.

3 International Energy Agency (IEA). (2021). *Empowering cities for a net zero future: Unlocking resilient, smart, sustainable urban energy systems.* Paris: OECD Publishing.

4 United Nations Environment Programme. (2024). *Global status report for buildings and construction.* Nairobi: UNEP.

5 Akyol, I.C., Halacli, E.G., Ucar, S., Iseri, O.K., Yavuz, F., Guney, D., Gursoy, F.E., Duran, A., Akgul, C.M., Kalkan, S. and Dino, I.G., 2025. Machine learning based prediction of long-term energy consumption and overheating under climate change impacts using urban building energy modeling. *Sustainable Cities and Society,* 130, p.106500.

1.3. Energy Renovation Strategies

Passive Systems

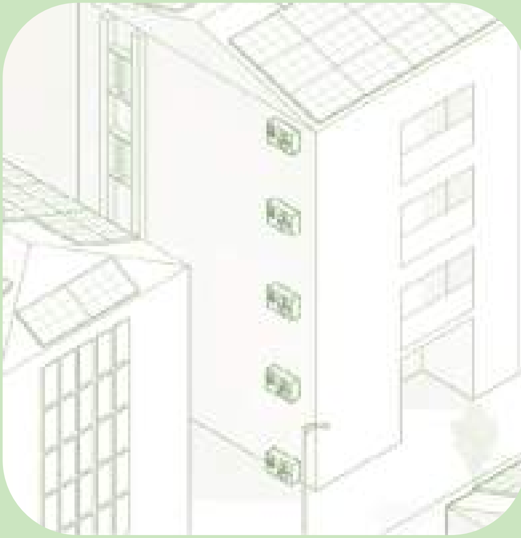
Envelope renovation (TS825 and Passivhaus)

Active Systems

Heat Pumps (HP)

Renewable Energy Systems

Photovoltaics (PV)

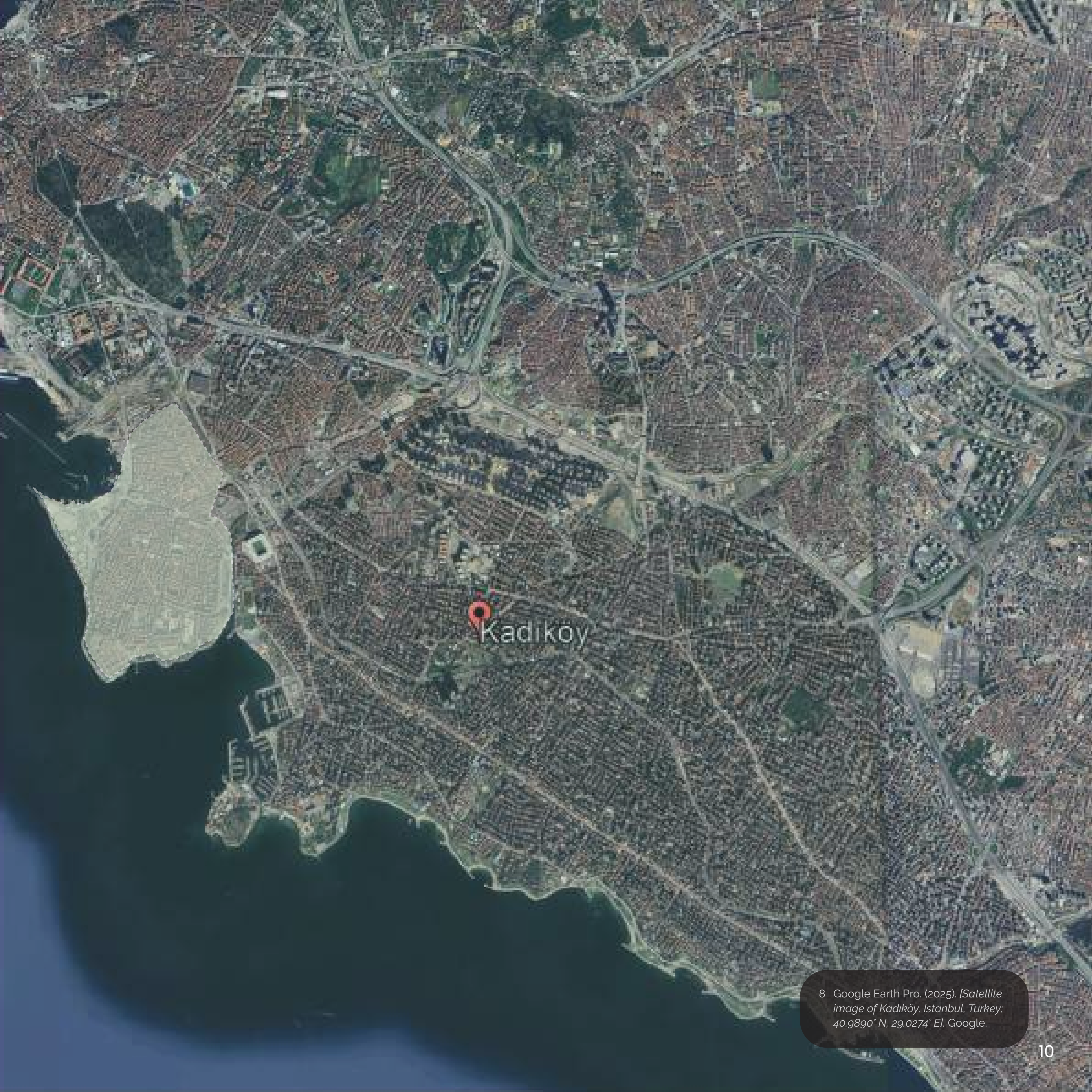


1.4. What İstanbul Aims with UP2030?

Istanbul is a densely populated city with **15.8 million inhabitants**. The city is experiencing severe heat waves intensified by climate change. Buildings with poor energy efficiency, particularly in low-income neighborhoods, have led to high energy consumption and associated CO_2 emissions. Due to traffic congestion and overcrowded public transport, the city also faces mobility problems. To address these challenges, Istanbul has aligned itself with the **C40 "Deadline 2020" initiative**, committing to making the city carbon neutral by 2050. In line with this commitment, Istanbul Metropolitan Municipality (IMM) published its **C40 Climate Change Action Plan (CAP)** in 2021, presenting its greenhouse gas emission inventory, future scenario analyses, targeted reduction rates by milestone years, and prioritised actions. In addition, Istanbul's Sustainable Urban Mobility Plan aims to overcome transport-related challenges, including greenhouse gas emissions. Through **UP2030**, Istanbul focuses on complementing these plans by deploying web based decision making tool that enables decision makers to analyse key performance indicators and negotiate trade-offs more effectively in the decision-making process for decarbonization of buildings through energy renovation strategies [6].

6 UP2030 Project. *Unique City: Istanbul.*

7 Google Earth Pro. (2025). *[Satellite image of Istanbul, Turkey; 41.0082° N, 28.9784° E].* Google.



8 Google Earth Pro. (2025). *[Satellite image of Kadıköy, Istanbul, Turkey; 40.9890° N, 29.0274° E]*. Google.

Kadıköy Pilot Area

Kadıköy is selected as a **pilot area** for urban building energy modeling and decarbonisation pathways. This neighborhood represents diverse building thermal characteristics due to the various construction years (1960s to now). The majority of the buildings date back to before 1980, indicating that the envelope materials and HVAC (Heating, Ventilation, and Air Conditioning) systems are mostly outdated. Therefore, there is a need for **efficient energy renovation strategies** to reduce energy consumption, lower CO_2 emissions, and ensure economic feasibility.

9 Google Earth Pro. (2025). [Satellite image of Caferağa, Kadıköy, Istanbul, Turkey; 40.9883° N, 29.0257° E]. Google.


Kadıköy

2. KADIKÖY

PILOT AREA

2.1. Demographic Status

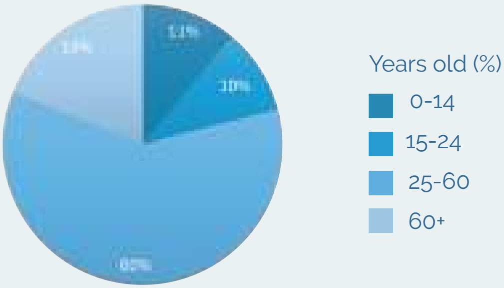
Kadıköy

It has a population of **485,233**, representing 3.06% of Istanbul's total population. The majority of its residents, around 60%, are between the ages of 25 and 64, categorized as the working-age population. In terms of gender distribution, women comprise 55% of Kadıköy's population.

Elementary families, consisting of two parents and their children, constitute the largest share at 38.23% within the household types. Over the years, there has been an increase in single-person and single-parent households. With **2.35 persons per household** Kadıköy has the smallest average household size among Istanbul's 39 districts

Kadıköy is rich in green spaces, featuring 102 parks and green areas. The district contains a total of 1,106,675 m^2 of active green space and 853,764 m^2 of passive green space. This corresponds to **2.28 m^2 of active green space per person** [10].

Population - Age Distribution

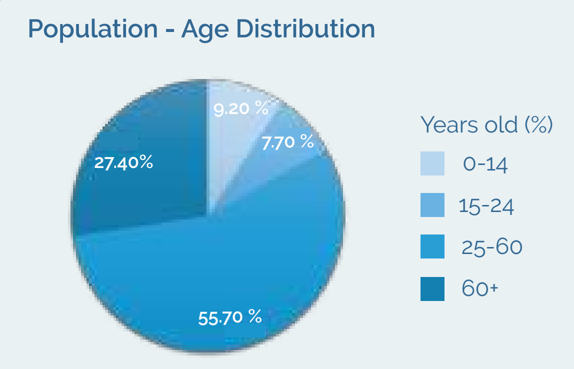


10 Kadıköy Municipality. (2021). *Kadıköy 2030 current situation report.* Istanbul: Kadıköy Municipality.

Pilot Area

The area is one of the oldest neighborhoods in Kadıköy, with a population of approximately **22,760**. The population is composed of 55.8% males and 44.2% females. Overall, the demographic structure is characterized by a predominance of middle-aged and elderly residents (25-60+), indicating an aging population profile.

The pilot area has the highest proportion of single-person households in Kadıköy, with about **9%** of residents living alone. Moreover, the area has a population density of approximately **18 people per 1,000 square meters**; consequently, high energy use intensity is present in the area [11].

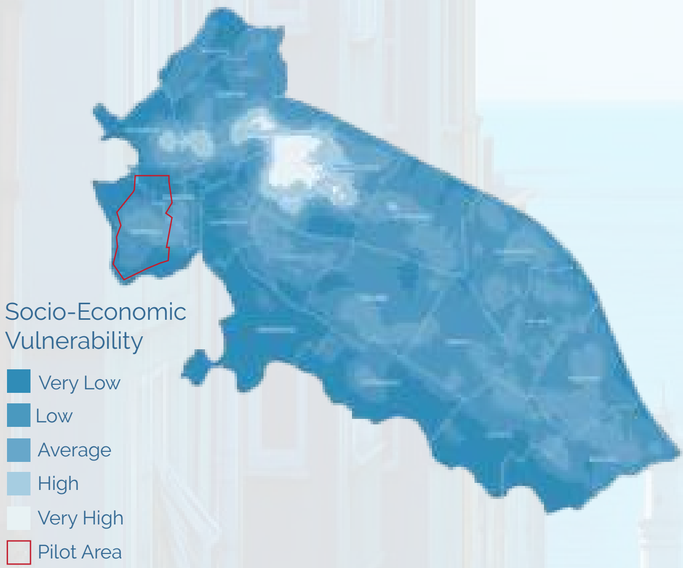


[11] Kadıköy Municipality. (2021). *Kadıköy 2030 current situation report.* Istanbul: Kadıköy Municipality.

2.2. Socio-Economic Vulnerability

According to the **Kadıköy 2030 Current Situation Report**, socio-economic vulnerability is higher in neighborhoods designated as **Urban Transformation Areas** and their surrounding zones. In contrast, **coastal neighborhoods** exhibit lower levels of vulnerability. Within the project's pilot area, vulnerability remains high or average, primarily due to the high proportion of residents aged 65 and above [12].

Kadıköy Socio-Economic Vulnerability Map



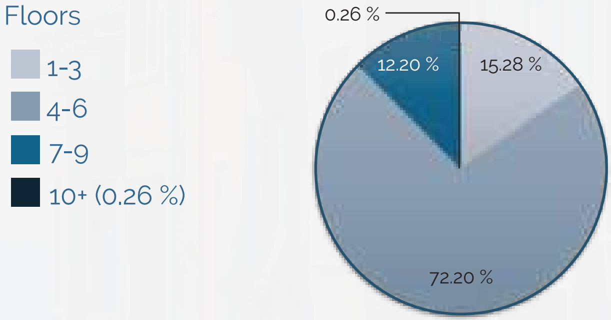
12 Kadıköy Municipality. (2021). *Kadıköy 2030 current situation report.* Istanbul: Kadıköy Municipality.

13 Gazete Kadıköy. (2017). *Yeldeğirmeni kitap oldu.*

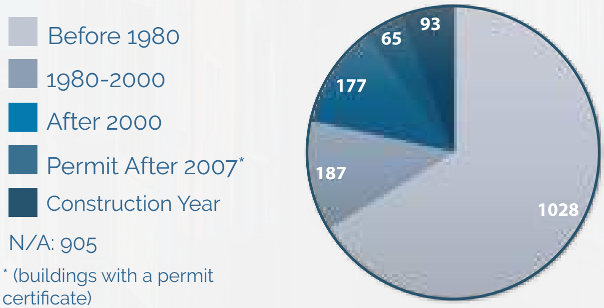
2.3. The Existing Building Stock

The pilot area consists of **2,455 buildings** and **12,016 zones** including residential units and offices, most of which are **high-density, mid-rise structures** typically between four and six storeys. The majority were built before 1980, reflecting an aging building stock that leads to **poor overall energy performance** in the neighborhood [14].

Number of Floors



Building Years



* (buildings with a permit certificate)

14 Kadıköy Municipality. (2021). *Kadıköy 2030 current situation report.* Istanbul: Kadıköy Municipality.

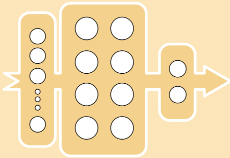
3. UBEM:

AI-POWERED DECISION MAKING TOOL FOR DECARBONIZATION

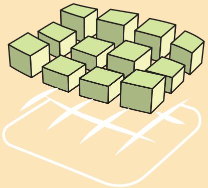
3.1. Key Enabling Technologies

Key enabling technologies such as **Urban Building Energy Modeling (UBEM), Artificial Intelligence (AI),** and **Geographical Information Systems (GIS)** play a central role in clean energy transition planning. They enable the evaluation of energy efficiency, renewable energy integration, and economic feasibility while the calculation and visualization of KPIs enhance the decision-making processes.

UBEM



AI



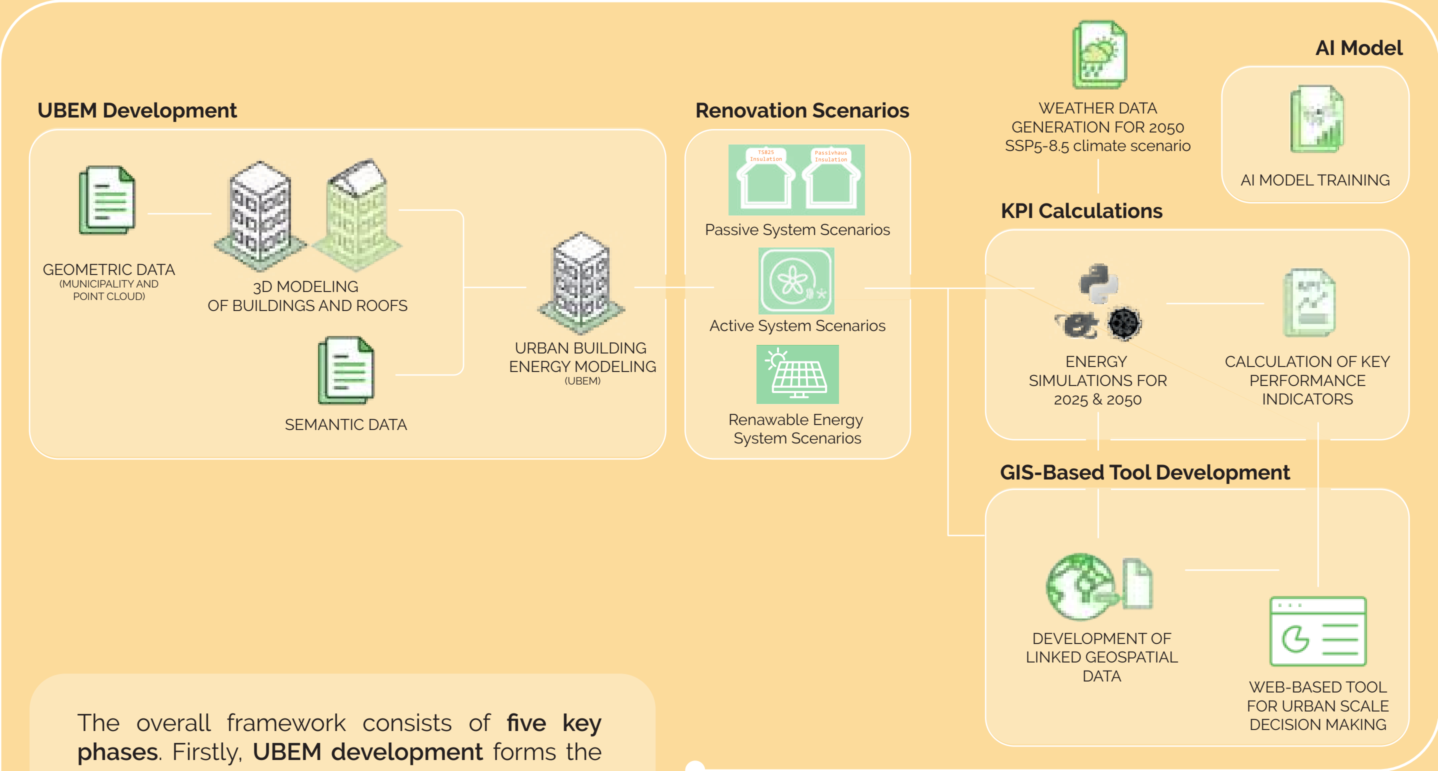
GIS

Urban Building Energy Modeling (UBEM)

UBEM is a powerful tool for understanding both buildings' **individual performance** and their **cumulative energy behavior** at the urban scale. It allows the analysis of energy demand patterns and the evaluation of future scenarios for informed urban planning and energy policy decisions. **UBEM-based simulations** estimate energy use, peak loads, and potential savings from energy renovations. Also, simulations helps planners and policymakers to design approaches to reduce carbon emissions and optimize energy use in changing climatic conditions.

While **decision-making processes** that focus on single-building energy renovation are common, these processes remain ineffective in delivering **wide-scale impact**, as they miss the opportunity to coordinate renovation scenarios, optimize the use of economic resources, and address cities' broader sustainability goals.

The Overall Framework



The overall framework consists of **five key phases**. Firstly, **UBEM development** forms the basis for the other phases. Following, **energy renovation scenarios** are developed in three main categories: *passive systems, active systems, and renewable energy systems scenarios.* Based on these scenarios, **UBEM-based simulations** are performed for the years 2025 and 2050, and the corresponding **KPIs are calculated**. In addition, **an AI model is trained** using the obtained data, capable of rapidly and accurately estimating energy consumption. Finally, all building data, calculated KPIs, and geospatial data are integrated into the **AI-powered decision making tool** and presented in a clear and visual format.

UBEM Development

UBEM development mainly relies on two data layers: ***a 3D geometric model and semantic data.***

3D Modeling

The **3D modeling** of the pilot area consists of two phases: *floor mass modeling and roof modeling*. The 3D building geometries are acquired from the city **GIS database**, buildings' **Energy Performance Certificates (EPCs)**, and the **National Address Inquiry System**. In total, **2,455 buildings and 12,016 thermal zones** are modeled and simulated. Each floor is modeled as a separate thermal zone in the EnergyPlus simulation software.

Semantic Data

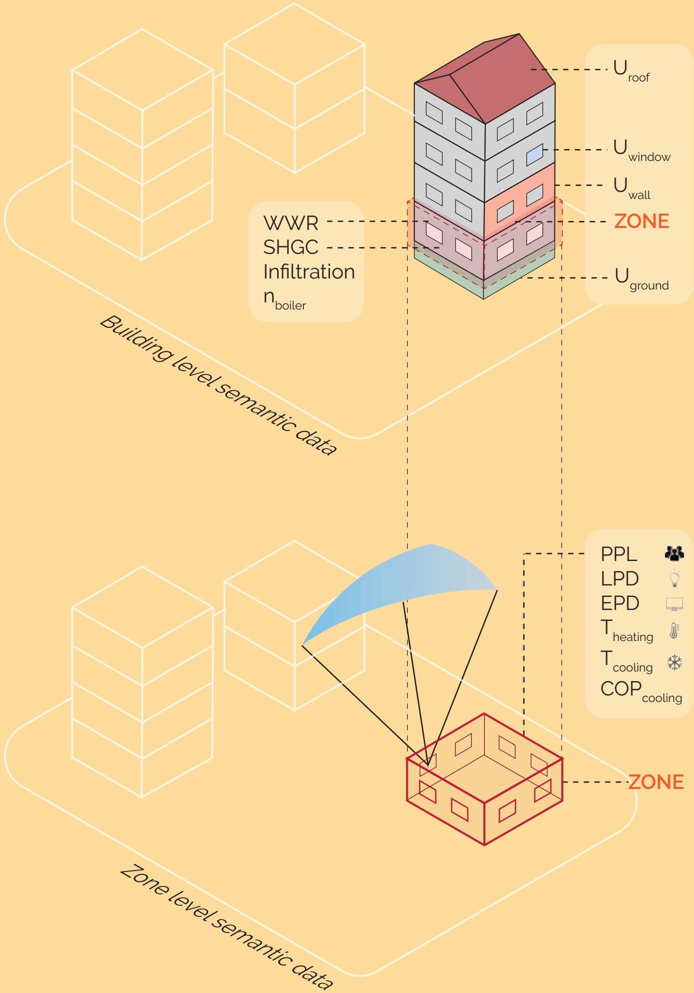
Semantic data refers to structured information that defines the **non-geometric attributes** of a building that are required in energy models. Semantic data can be divided into *building-level data* and *zone-level data.*

Building level semantic data

Building level data include **envelope materials** (U_{Roof}, U_{Wall}, U_{Window}, U_{Ground}), **boiler efficiency** (η_{boiler}), **window-wall ratio** (WWR) in four directions, **solar heat gain coefficient** (SHGC), and facade **infiltration** rate.

Zone level semantic data

Zone-level data varies from zone to zone, including internal loads including **people density** (PPL), **lighting power density** (LPD), **equipment power density** (EPD), **heating setpoint** ($T_{Heating}$), **cooling setpoint** ($T_{Cooling}$), and **air conditioning efficiency** ($COP_{cooling}$).



3.2. Development of Energy Renovation Scenarios

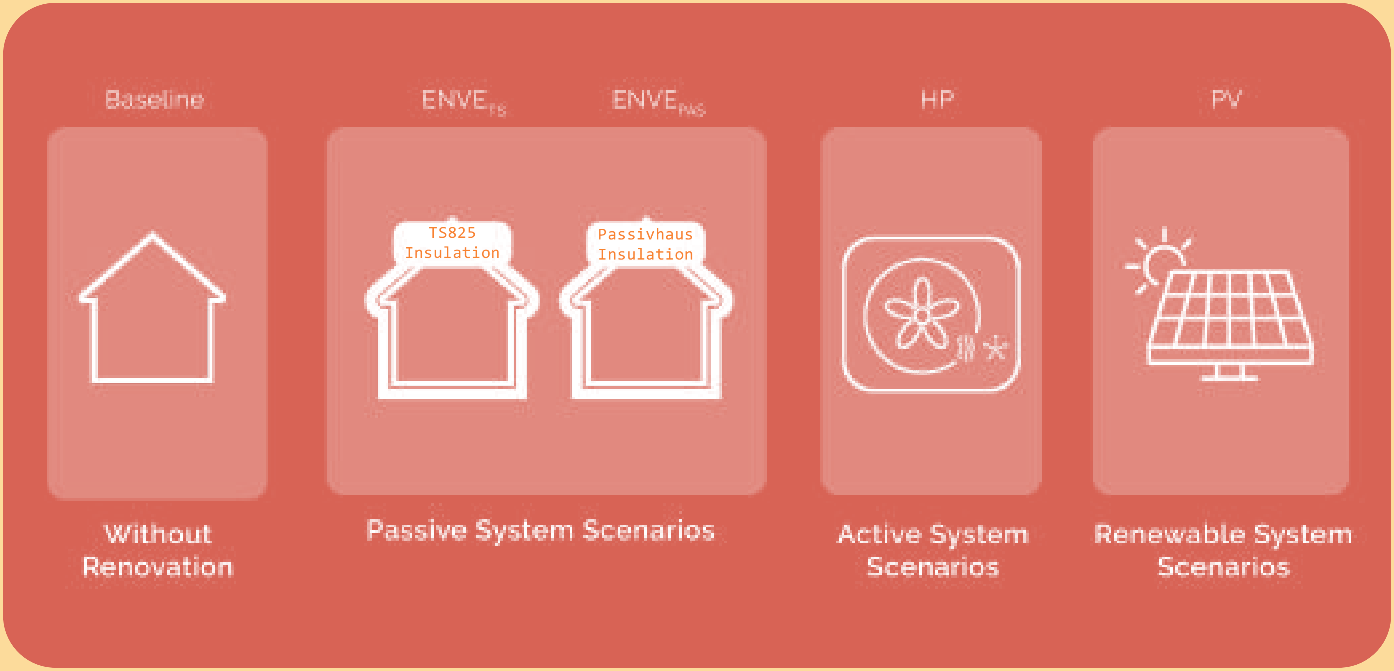
In this phase, the **energy renovation scenarios** are developed, and their effectiveness is evaluated using the KPIs (section 3.4.). There are three main categories of energy renovation scenarios that aim to achieve the following:

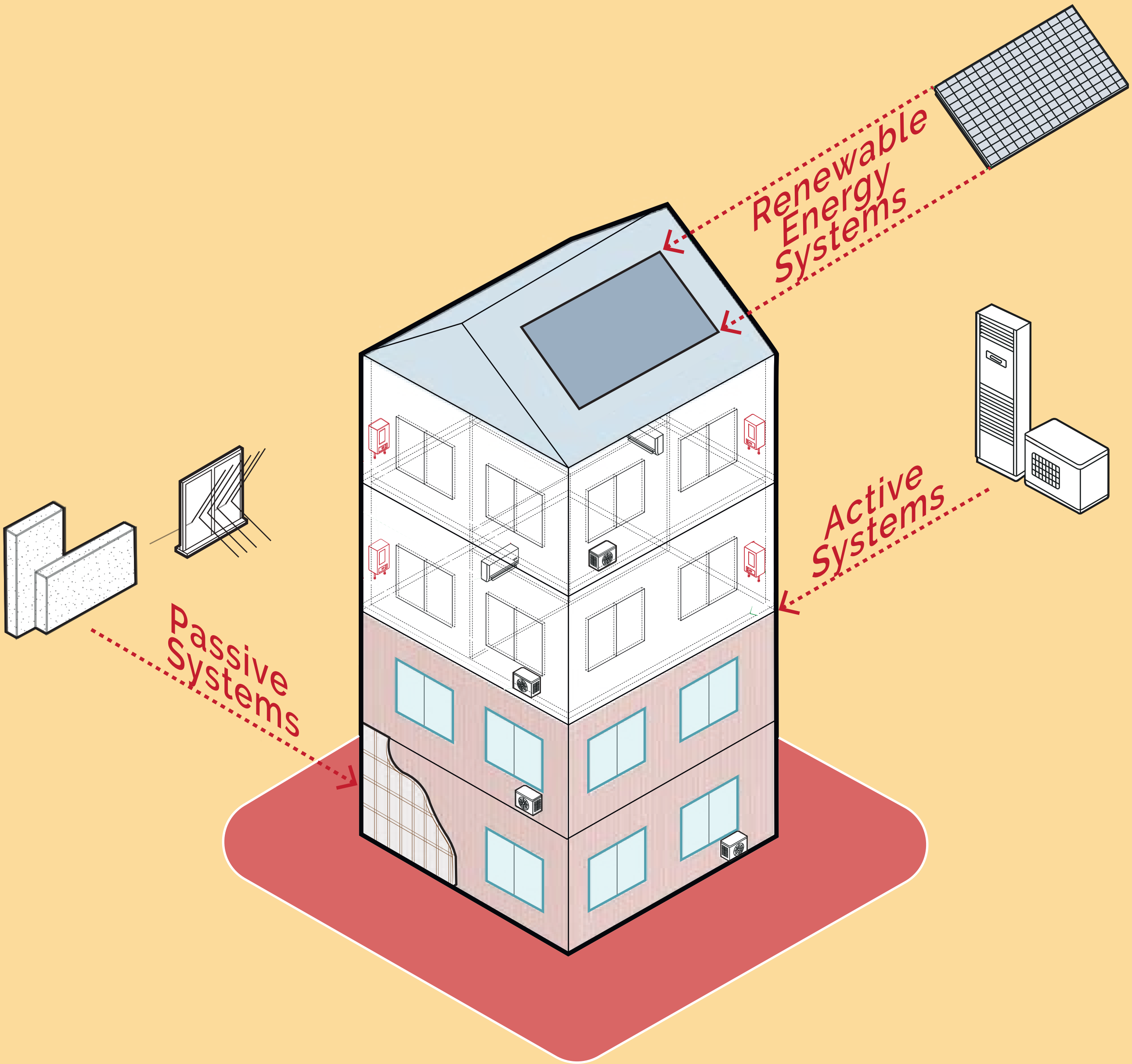
-Reducing energy use with passive and active systems,
-Transitioning to high-efficiency, electrified active energy systems,
-Generating on-site energy from renewable systems.

Passive system scenarios: Based on the Turkish Thermal Insulation Requirements for Buildings (TS825) and the Passivhaus standards, these scenarios enhance envelope insulation to minimize heating and cooling loads.

Active system scenarios: Introduce heat pump (HP) technologies for both heating and cooling, enabling electrification and supporting the transition from fossil fuels to renewable energy.

Renewable energy system scenarios: Evaluate rooftop photovoltaic (PV) potential to meet buildings' electricity consumption.




Renewable
Energy
Systems
Active
Systems
Passive
Systems

Categories of Renovation Scenarios

Code	Envelope Retrofit	Heat Pump	Photovoltaic Panel	
Baseline	X	X	X	Category Scenarios
$ENVE_{TS}$	TS825	X	X	
$ENVE_{PAS}$	Passive House	X	X	
HP	X	✓	X	
PV	X	X	✓	
$ENVE_{TS}$ + HP	TS825	✓	X	Combined Scenarios
$ENVE_{PAS}$ + HP	Passive House	✓	X	
HP + PV	X	✓	✓	
$ENVE_{TS}$ + PV	TS825	X	✓	
$ENVE_{PAS}$ + PV	Passive House	X	✓	
$ENVE_{TS}$ + HP + PV	TS825	✓	✓	
$ENVE_{PAS}$ + HP + PV	Passive House	✓	✓	

The proposed energy renovation strategies aim to (i) reduce heating energy use, (ii) reduce cooling energy use, (iii) mitigate indoor overheating degree (IOD), (iv) transition from low-efficiency boilers to heat pumps to support decarbonization, and (v) promote clean energy transition through the integration of photovoltaics (PV). In line with these objectives, several energy renovation scenarios have been evaluated:

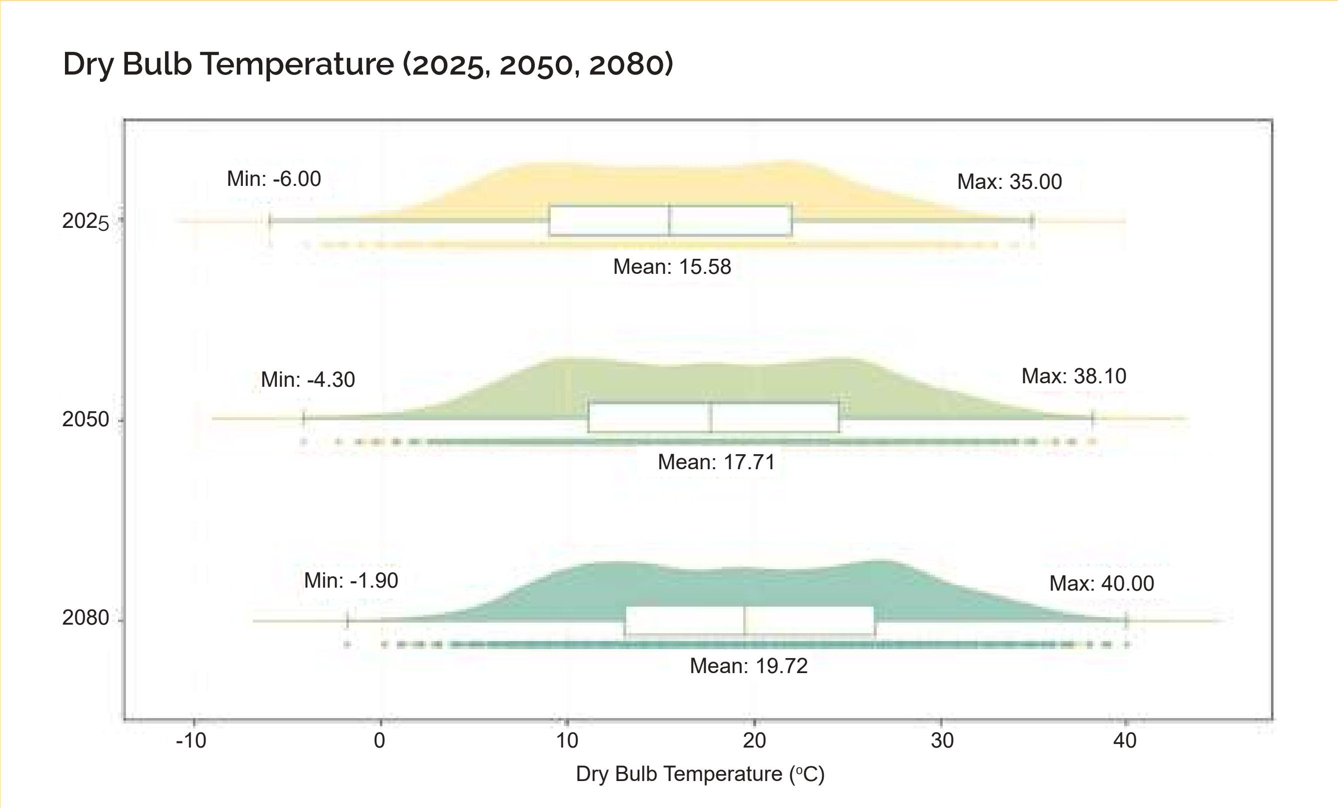
Baseline, representing the current building stock;
$ENVE_{TS}$, where insulation is improved (U_{wall}, U_{roof}, U_{window}, U_{ground}) based on the Turkish "Thermal Insulation Requirements for Buildings" (TS825) standard;
$ENVE_{PAS}$, where insulation is improved (U_{wall}, U_{roof}, U_{window}, U_{ground}) based on Passivhaus standard;
HP, where natural gas boilers are replaced with heat pumps for both heating and cooling;
$ENVE_{TS}$+HP, where heat pumps are combined with the $ENVE_{TS}$ scenario;
$ENVE_{PAS}$+HP, where heat pumps are combined with the $ENVE_{PAS}$ scenario.

In addition, the scenarios **$ENVE_{TS}$+PV, $ENVE_{PAS}$+PV, $ENVE_{TS}$+HP+PV**, and **$ENVE_{PAS}$+HP+PV** denote the same strategies as above but with the inclusion of PV systems to further enhance the clean energy transition.

3.3. Future Weather File Generation

The **Typical Meteorological Year (TMY) file** of Istanbul, which is based on the period 2007-2021, is used to perform baseline simulations representing current weather conditions (W2025). Then, two weather files (W2050 and W2080) are generated for 2050 and 2080 climatic conditions using the **'Future Weather Generator'**. It is used to create future weather files based on the morphing method and the latest climate data model (EC-Earth3) [15]. Four Shared Socioeconomic Pathways (SSP), including SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5, are implemented in the tool.

The **SSP5-8.5 scenario** is selected as it represents a future climate with intensive fossil fuel-driven development, leading to very high greenhouse gas emissions and severe climate impacts. Below, the graph shows the distribution of dry bulb temperatures for the years 2025, 2050, and 2080 in Istanbul, illustrating the projected impacts of climate change.



15 Rodrigues, E., Fernandes, M. S., & Carvalho, D. (2023). Future weather generator for building performance research: An open-source morphing tool and an application. *Building and Environment,* 230, 109123.

3.4. Key Performance Indicators

Energy Simulations

2455 buildings for each year (2025 and 2050) are modeled and simulated using EnergyPlus at **annual, monthly, and hourly** time steps. The models are calibrated using the buildings' Energy Performance Certificates (EPCs) and actual natural gas consumption bills. Each renovation scenario is modeled and simulated separately.

The Key Performance Indicators are classified into three main categories: **Environmental, Economic, and Thermal Comfort** indicators. **Environmental indicators** serve as critical metrics for evaluating how much a building contributes to environmental impacts through its energy consumption and associated emissions. **Thermal comfort indicators** capture indoor thermal discomfort for the occupants. **Economic indicators** provide insight into the financial viability and the payback of the energy renovation scenarios.

Environmental Indicators

- Heating Energy Consumption
- Cooling Energy Consumption
- CO2 Reduction
- PV-Based Electiricity Production
- Electricity Self Sufficiency Index

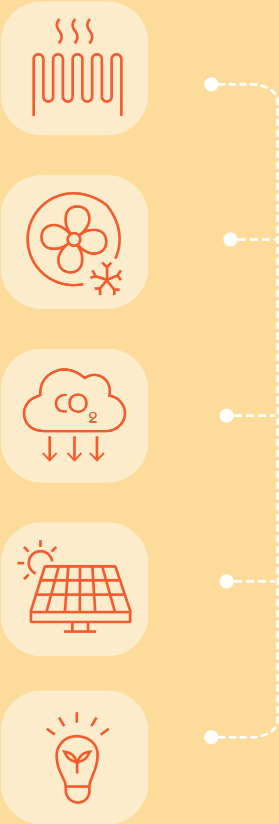
Economic Indicators

- Return On Investment
- Discounted Payback Period
- Net Present Value
- Abatement Cost

Thermal Comfort Indicators

- Indoor Overheating Degree

Environmental Indicators



Heating Energy Consumption (kWh/m²)

the amount of energy used to heat a space to an indoor setpoint temperature. It is calculated by taking into account the efficiency of the heating system.

Cooling Energy Consumption (kWh/m²)

the amount of energy used to cool a space to an indoor setpoint temperature It is calculated by taking into account the efficiency of the cooling system.

CO_2 Reduction (tCO_{2eq})

the decrease in carbon dioxide emissions achieved when renovation scenarios are applied to the building. In addition to heating and cooling consumptions, lighting and equipment electricity consumptions are also considered in the calculations.

PV-Based Electricity Production (kWh)

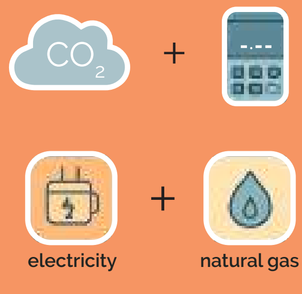
the amount of electricity generated through rooftop photovoltaic systems.

Electricity Self Sufficiency Index (%)

the ratio between PV-based electricity generation and building electricity energy demand.

Emission Calculations

Annual CO_2 emissions are calculated by multiplying the annual energy consumption of each source by its respective emission factor. For natural gas, fixed emission factors are adopted from international authorities such as the Intergovernmental Panel on Climate Change (IPCC). Electricity CO_2 emissions are quantified by multiplying end-use energy with emission factors for the current and future years, as obtained from ETKB (2024), and Enerdata (2025).



Thermal Comfort Indicator

Indoor Overheating Degree (ºC)

the extent and duration by which indoor temperatures exceed a defined comfort threshold, indicating thermal discomfort or potential health risk due to excessive indoor heat. IOD is calculated for the zones that rely only on natural ventilation.

Economic Indicators

Discounted Payback Period (year)

the period of time required for an investment to pay off, considering that future savings are valued slightly less over time.

Net Present Value (€)

the present value of the future cash flows generated by the investment, discounted to the present.

Return On Investment (%)

the percentage of return on a project based on the capital investment, and helps to compare different investment options.

Abatement Cost (€/tCO_2)

the cost-efficiency of carbon reductions across scenarios associated with reducing environmental impact, such as investing in energy efficiency renovations and reneawable energy integration.

Energy Tariffs

Households currently benefit from subsidized, inexpensive natural gas. Also, Türkiye implements tiered electricity pricing, where prices significantly increase after an annual consumption limit of 5,000 kWh is exceeded. These regulations pose a practical challenge for HP scenarios: when heating is completely met by electricity and cooling is subject to increase due to global warming, it becomes highly likely that households will surpass this threshold. Therefore, despite the environmental advantages of the electrification of heating systems, the tightening of tariff structures may complicate the economic feasibility of such initiatives.

4. AI-BASED MODELS

4.1. Artificial Intelligence (AI)

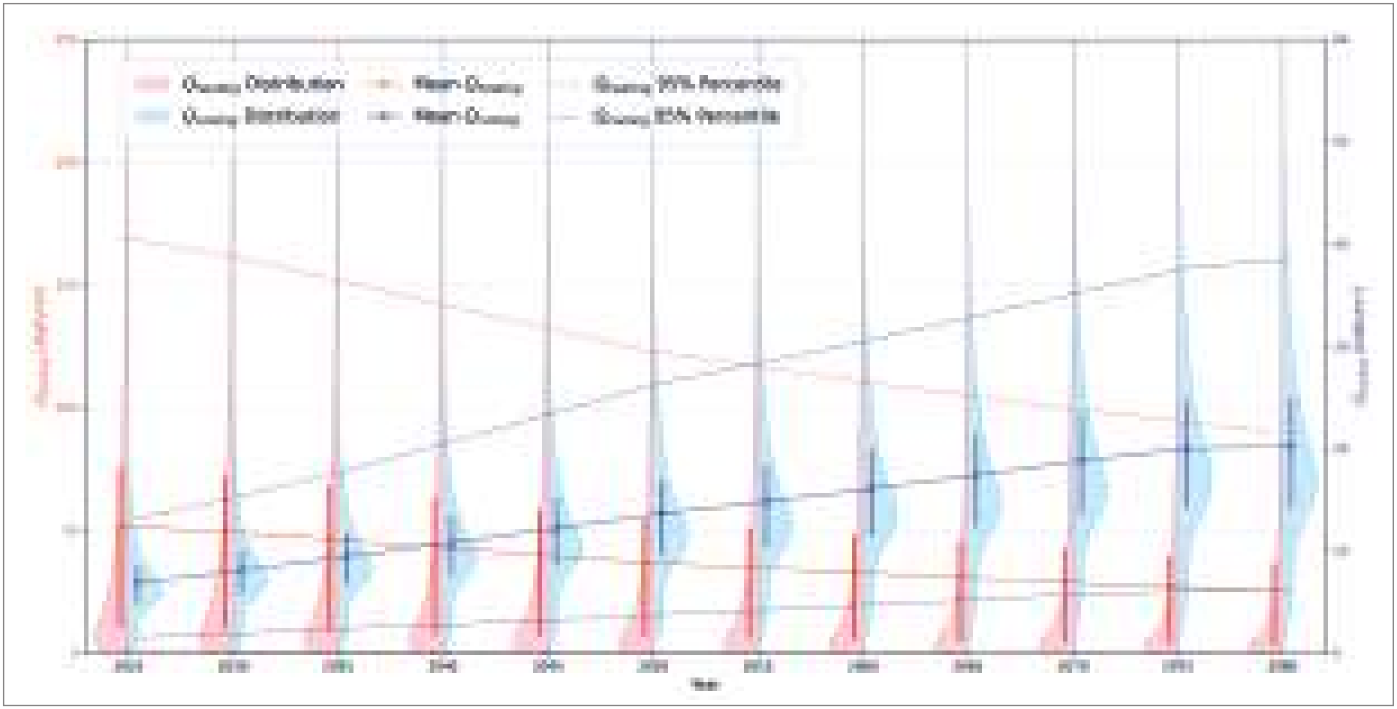
UBEM can support the evaluation of building energy performance across the urban scale. However, the use of UBEM is constrained by: (i) High modeling effort and the need for modeling expertise, and (ii) High computational demand. Alternatively, UBEM-assisted AI-based approaches can provide rapid and reliable predictions by capturing complex patterns and dependencies between input parameters and output variables. As such, AI models can support the decision-making process for building energy renovation planning, providing fast, scalable, and robust prediction capabilities.

Multi-Layer Perceptrons (MLPs)

Multi-Layer Perceptrons (MLPs) are artificial neural networks, known for their ability to **approximate any continuous function**, making them suitable for regression problems, particularly in the prediction of building energy performance. **MLP-based models** are trained on a comprehensive dataset generated through UBEMs, allowing them to predict key performance indicators, including *heating energy consumption, cooling energy consumption,* and *IOD* for both residential and office zones. These models considered the long-term effects of climate change, capturing how rising temperatures affect energy consumption and the risk of indoor overheating. The developed MLP model not only performs discrete predictions for 2020, 2050, and 2080, but also enables **continuous estimation of KPIs** for the intermediate years.

The MLP model demonstrates high predictive capability with high R^2 values above 0.98 for heating, cooling, and indoor thermal comfort. The model also provides continuous projections in yearly intervals between 2025 and 2080. The results showed realistic trends: heating demand decreased gradually and across the years, while cooling demand increased consistently until 2080. Indoor thermal comfort also exhibited an upward trend, similar to cooling, rising from 2025 to 2080 due to the impact of increasing outdoor temperatures.

The model's ability to capture both gradual and complex changes demonstrates its robustness for long-term applications. By providing predictions future climate conditions, the model enables comprehensive and rapid city-scale assessments. Such capacity is crucial for planners and policymakers, enabling them to foresee trends in energy demand and indoor comfort, and to assess strategies aimed at enhancing resilience, reducing carbon emissions, and supporting sustainable energy systems.



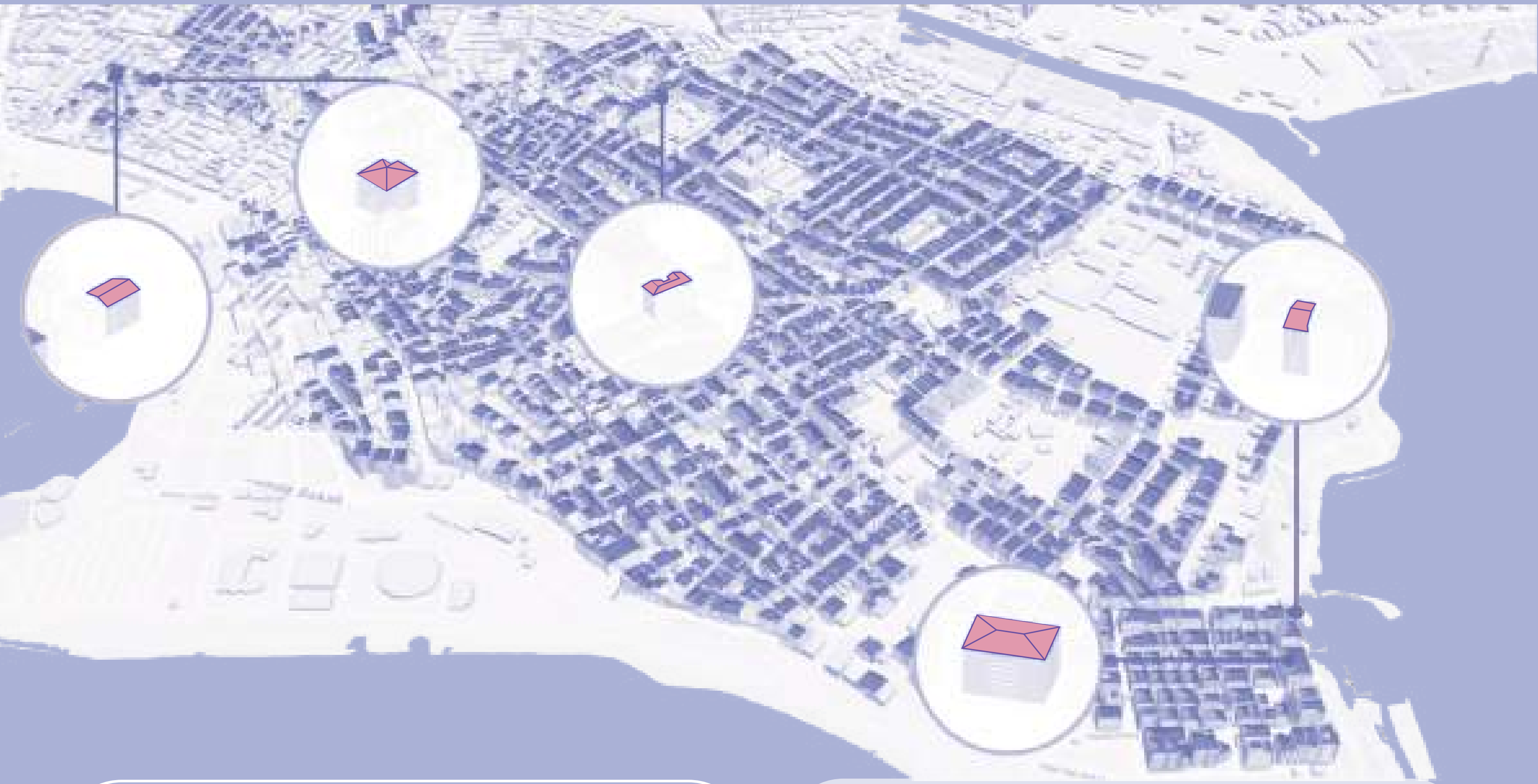
5 PHOTO-VOLTAICS

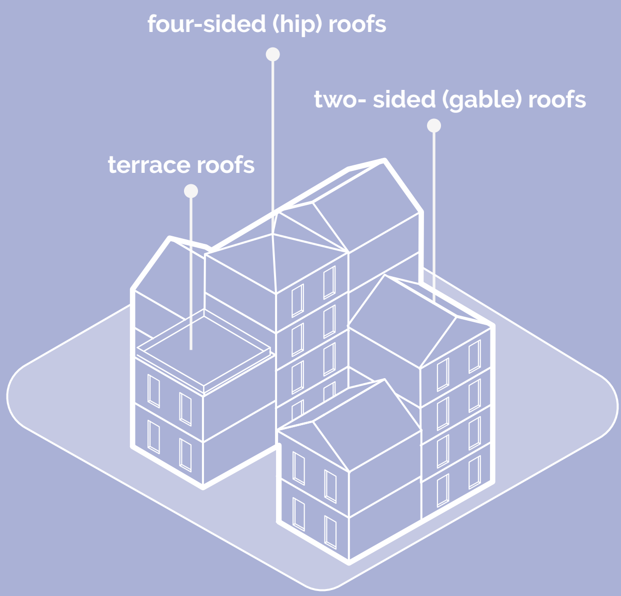
5.1.Roof Modelling

Photovoltaic systems play a crucial role in accelerating the clean energy transition. At the urban scale, rooftop photovoltaic (RTPV) systems are a promising energy source due to their high exposure to solar radiation, wide PV potential area, and ease of accessibility. **Rooftop modelling** enables the accurate estimation of solar photovoltaic (PV) potential across urban areas. Detailed roof modelling allows for the precise characterization of roof geometries, orientations, tilts, and shading effects, which are key determinants of solar irradiance and energy yield. By integrating roof modelling into UBEMs, cities can identify optimal roof surfaces for PV deployment and quantify the **achievable renewable energy generation**. Thus, roof modelling serves as a foundational step for data-driven planning and policy-making, linking the physical characteristics of the built environment with broader goals of energy decarbonization and climate resilience.

The roofscape of Kadıköy presents a diverse and dynamic urban texture, that reflects the geometric diversity of the neighborhood. The majority of the buildings have two-sided (gable) or four-sided (hip) roofs. More complex, multi-span roof structures are also scattered throughout the neighborhood, particularly on larger buildings or those with intricate layouts. In addition, some buildings feature flat terrace roofs.

16 teksomolika. (2020). Cityscape Istanbul Turkey photo from the birds eye view [Photograph]. 123RF.





The rooftop modeling process for buildings in the pilot area combined multiple data sources to ensure maximum accuracy and completeness, depending on the availability and quality of the data. The goal was to generate precise 3D rooftop geometries to support spatial and urban energy analyses. For this purpose, **LiDAR data** and **high-resolution satellite imagery** are obtained from the İstanbul Municipality. Where LiDAR data is available it is processed to produce high-accuracy models of **384 buildings.**

For the remaining buildings, high-resolution Google satellite imagery was used to reconstruct the roof surfaces and complete the 3D model.

Together, these datasets enabled the accurate modeling of all 2,455 buildings, providing a reliable foundation for assessing **rooftop PV potential** across the pilot area.

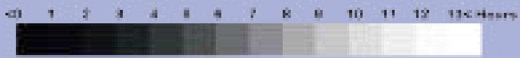
5.2. Solar Energy Analysis

The annual solar energy potential of each roof surface in the pilot area is calculated and visualized. Roof surfaces with darker tones indicates higher solar exposure, which can reach up to 107,760 kWh. A high annual solar yield indicates substantial solar exposure and the associated energy generation potential which can significantly help powering the electricity systems in buildings.

5.3. Sunlight Hours Analysis

September 21

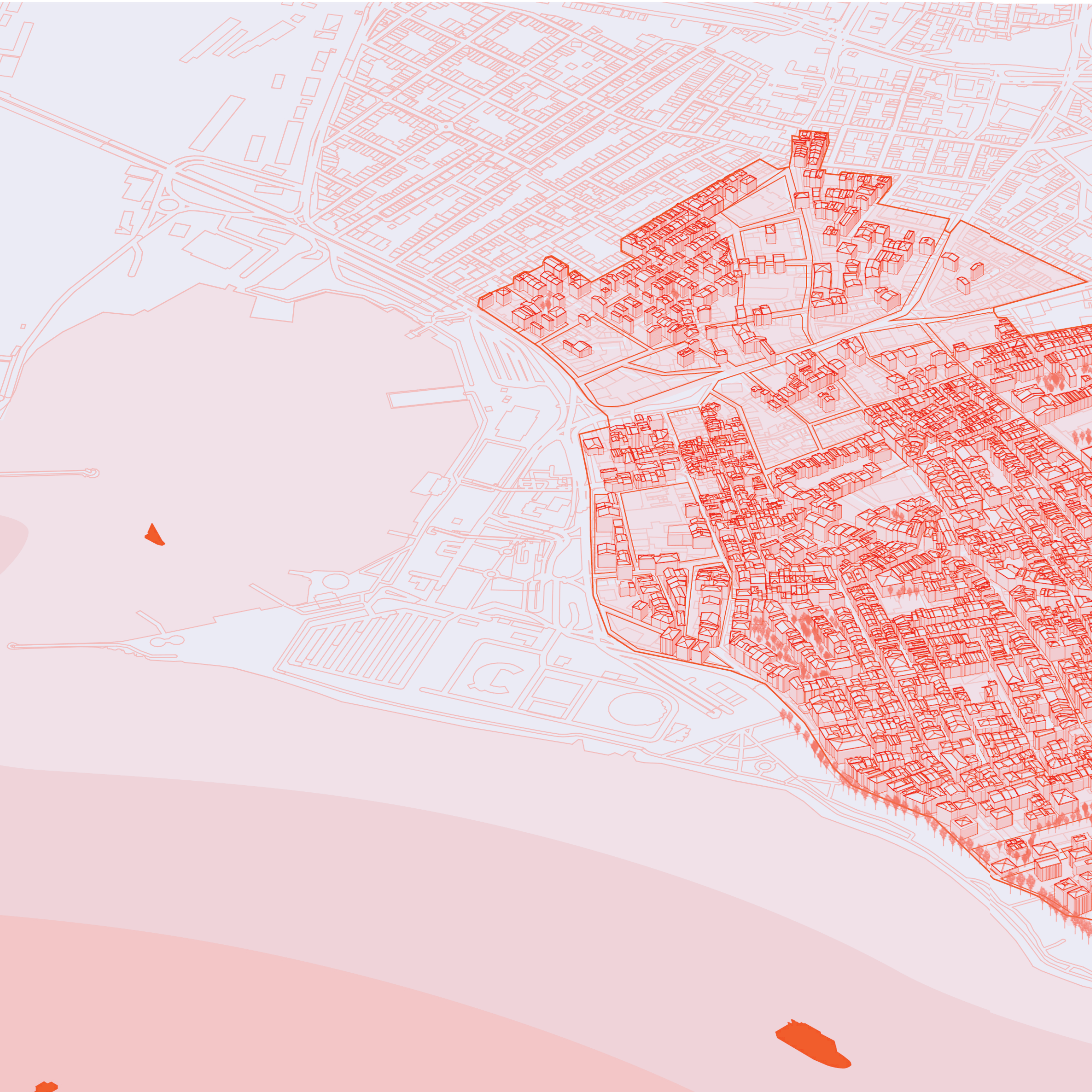
December 21



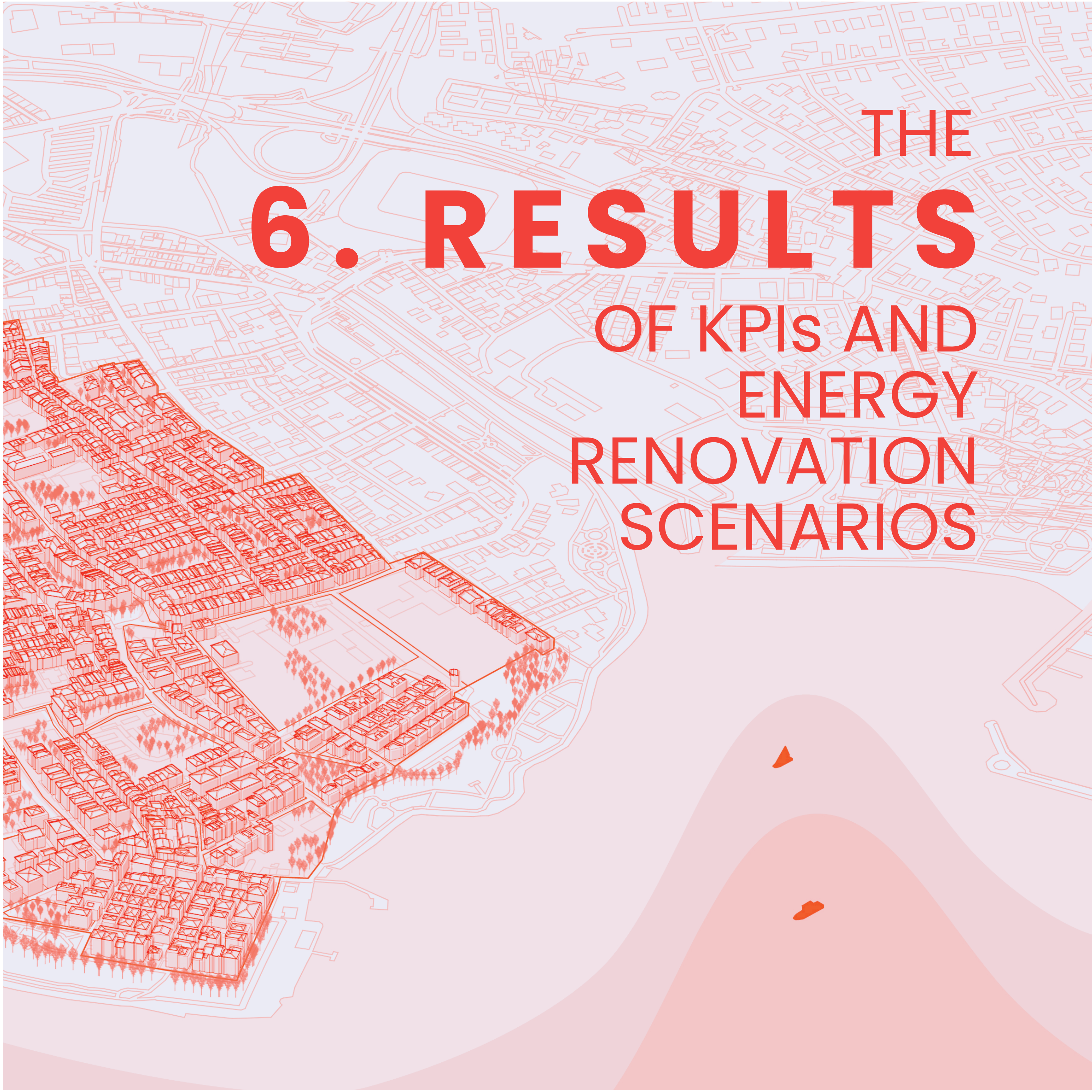
March 21

June 21

The sunlight analysis of the pilot area is based on four representative days of the year. The results show seasonal variations throughout the year, with significant sunlight exposure differences between summer and winter.

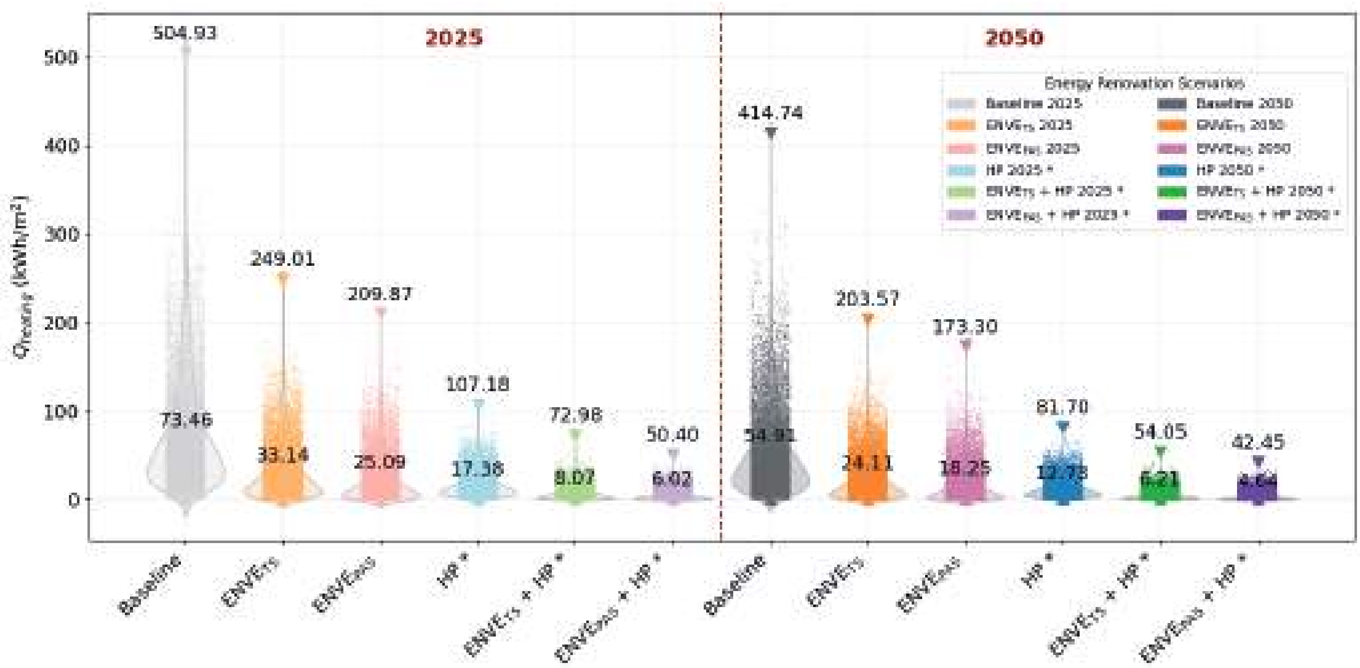


THE 6. RESULTS OF KPIs AND ENERGY RENOVATION SCENARIOS



6.1. The Results of Environmental Indicators and Energy Renovation Strategies

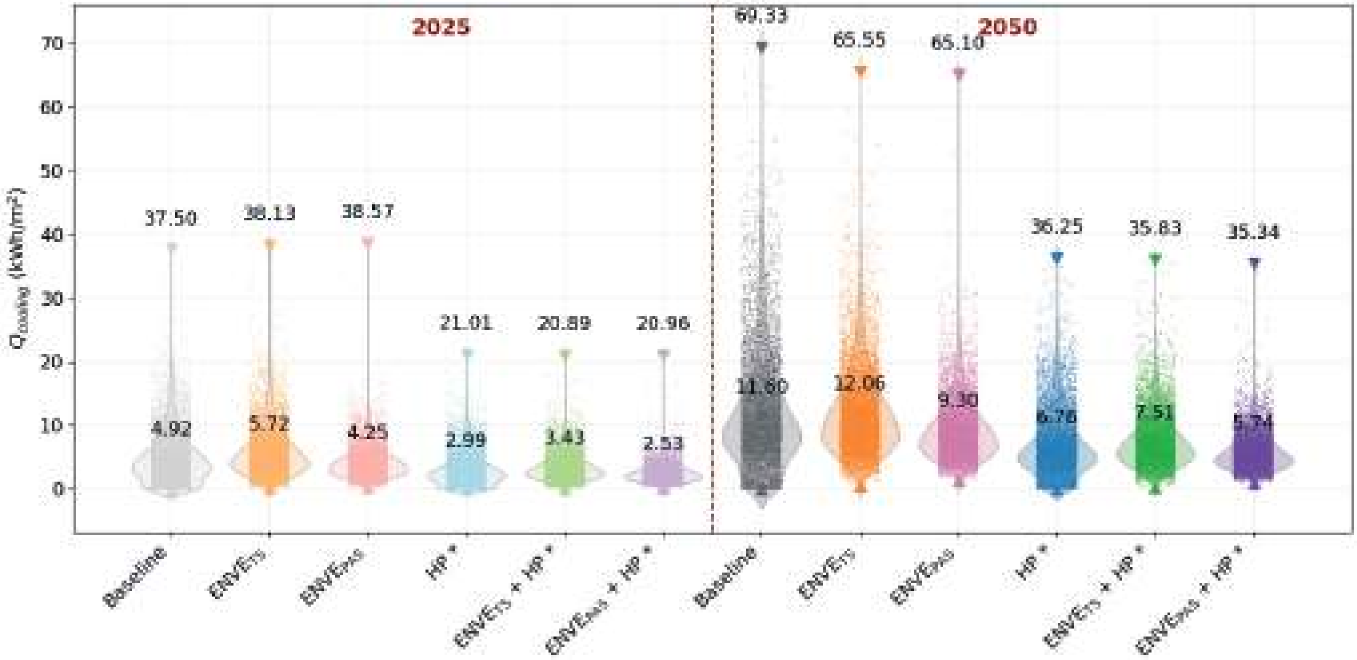
Heating Energy Consumption



Heating demand represents the **largest share of total energy consumption** in all buildings in the neighborhood.

In heating energy use, there is a consistent downward trend from 2025 to 2050 due to global warming. This corresponds to a reduction of approximately **25%**. A closer look into the renovation scenarios shows that heating energy use can be reduced up to threefold through envelope improvements, primarily lowering natural gas consumption. In contrast, the integration of heat pumps shifted the demand from natural gas to electricity and reduced heating energy use by up to **77%** across all scenarios, owing to heat pumps' higher coefficient of performance (COP) compared to the efficiency of natural gas boilers.

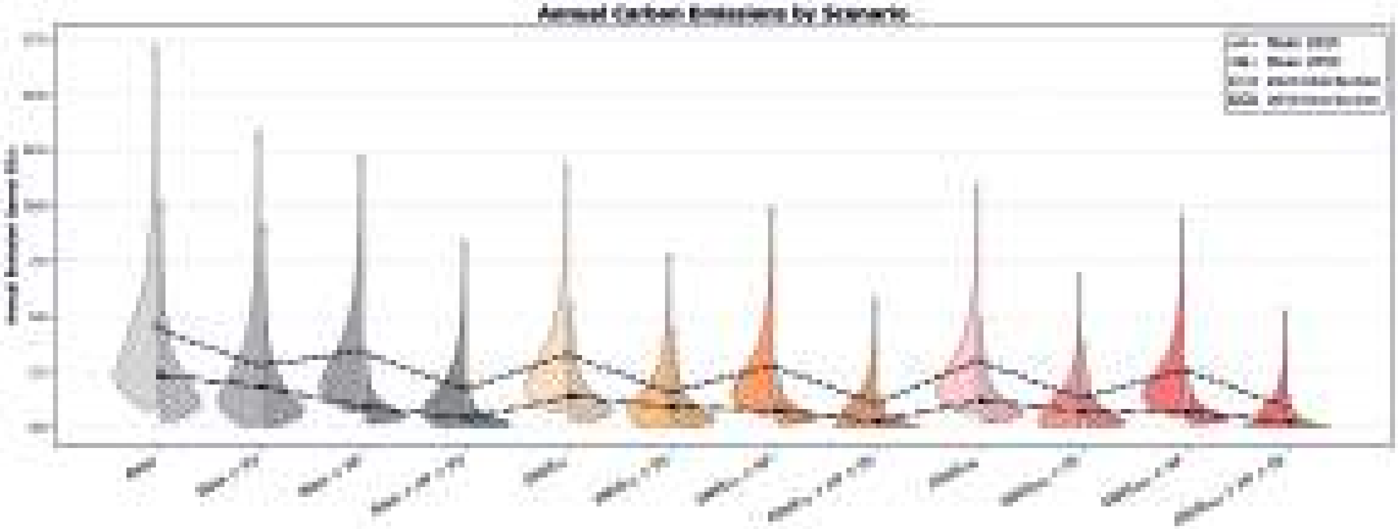
Cooling Energy Consumption

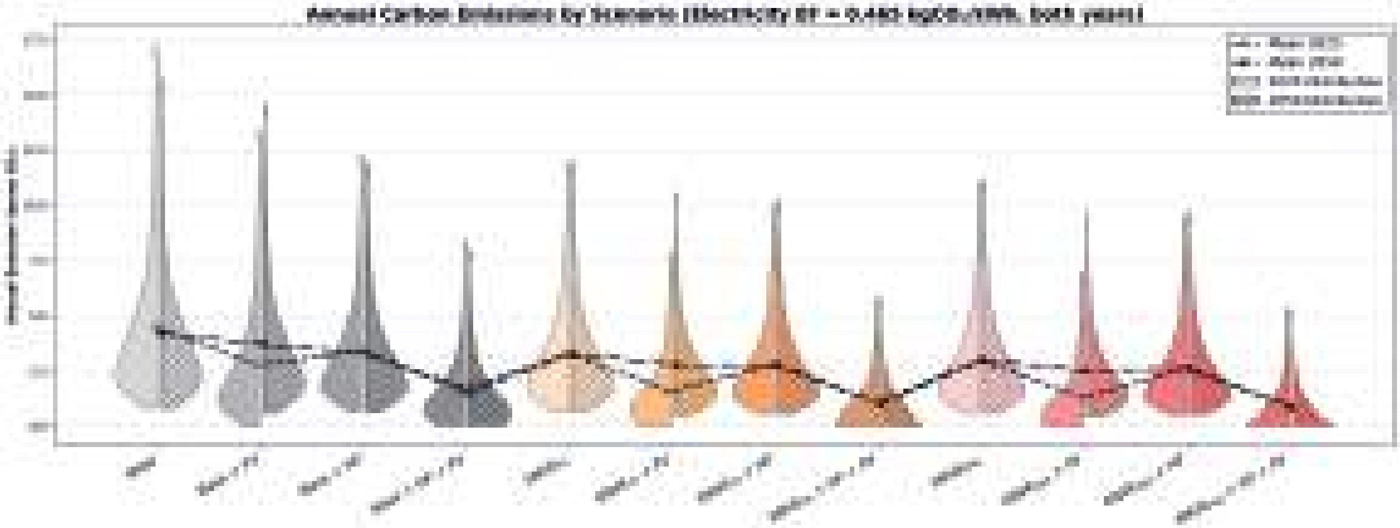


In contrast, the cooling energy use increased by approximately **2.3 times** across all scenarios from 2025 to 2050 due to the increasing temperatures. The cooling demand gradually declined **22%** from TS to Passivhaus scenarios.

Heat pumps are more efficient than conventional air conditioners and can reduce cooling energy use by approximately **40%**. Besides electrification and its potential to be coupled with PV, heat pumps have the additional benefit of meeting both heating and cooling loads in one single piece of equipment.

CO_2 Reduction





The analysis of CO2 emission distributions shows how each renovation scenario changes the environmental impact. The Baseline scenario has the highest and widest distribution of emissions, with mean values around 4–5 tonnes in 2025 and nearly 2–3 tonnes in 2050 (higher with the constant-EF). Introducing Heat Pumps reduces the mean, while adding Photovoltaics lowers both the median and the upper tails, bringing the mean to roughly 1.5–3 tonnes. Envelope renovations further reduce emissions; TS with Heat Pumps delivers additional reductions, and the TS+HP+ PV drives the mean sharply downward to about 1 tonnes.Passivhaus renovations push emissions toward the lowest levels, and coupling them with PV provides consistently nearly zero values, with Passivhaus envelope + HP + PV frequently falling below 1 tonne CO_2. Overall, if the expected decrease in the electricity emission factor (EF) is realized, electrification will lead to significant reductions in overall emissions. However, if this decline does not occur and the EF remains constant as a worst-case scenario, the increasing electricity demand projected for 2050 could result in higher total emissions compared to 2025.

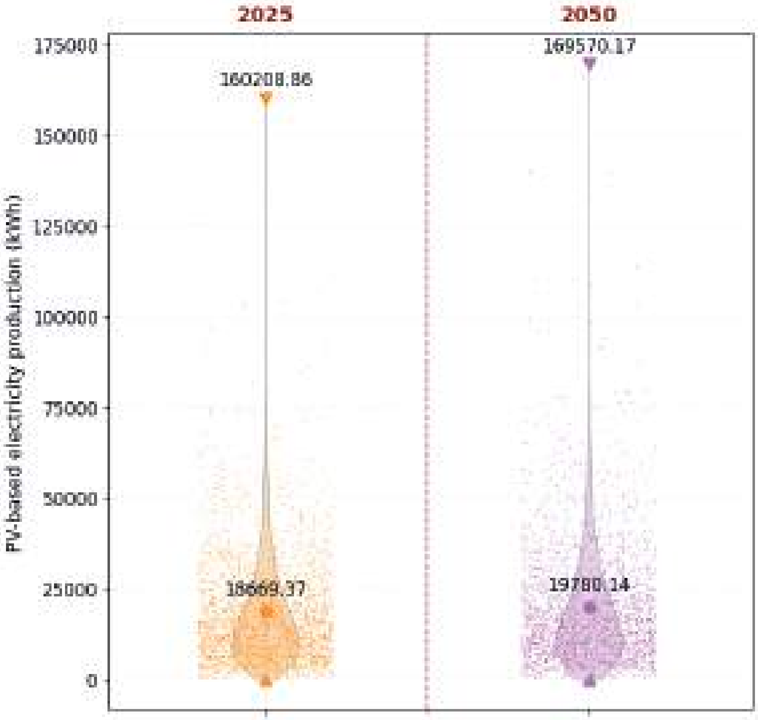
Key Takeaways

• Without improvement in the grid emission factor of electricity, emissions in 2050 will be equal to today's emissions.

• With constant grid emission factors, rising energy consumption is projected to exceed the contribution of photovoltaics, resulting in higher emissions.

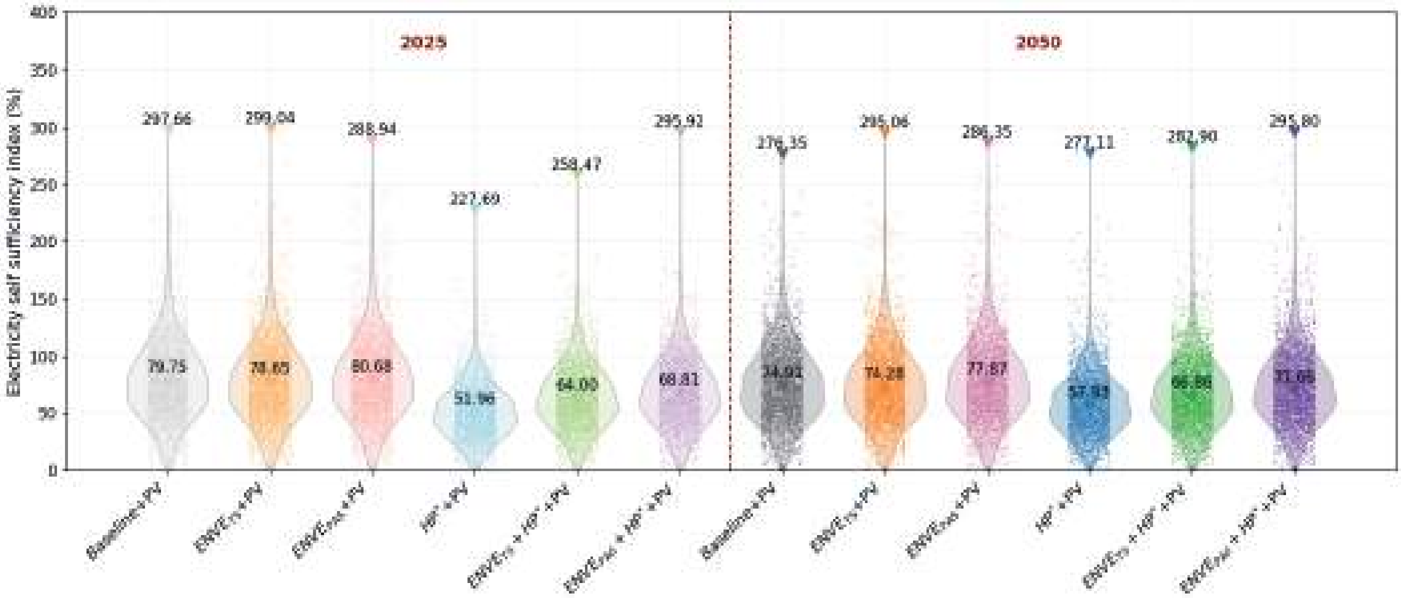
• To reduce the electricity emission factor, policy implementations must promote a cleaner grid by increasing the share of renewable energy sources.

PV-Based Electiricity Production



A comparative analysis of PV-based electricity production indicates a projected 6% increase between 2025 and 2050, primarily driven by rising solar radiation levels. This upward trend suggests that future building energy systems could increasingly benefit from enhanced on-site renewable energy generation.

Electricity Self Sufficiency Index by Scenario

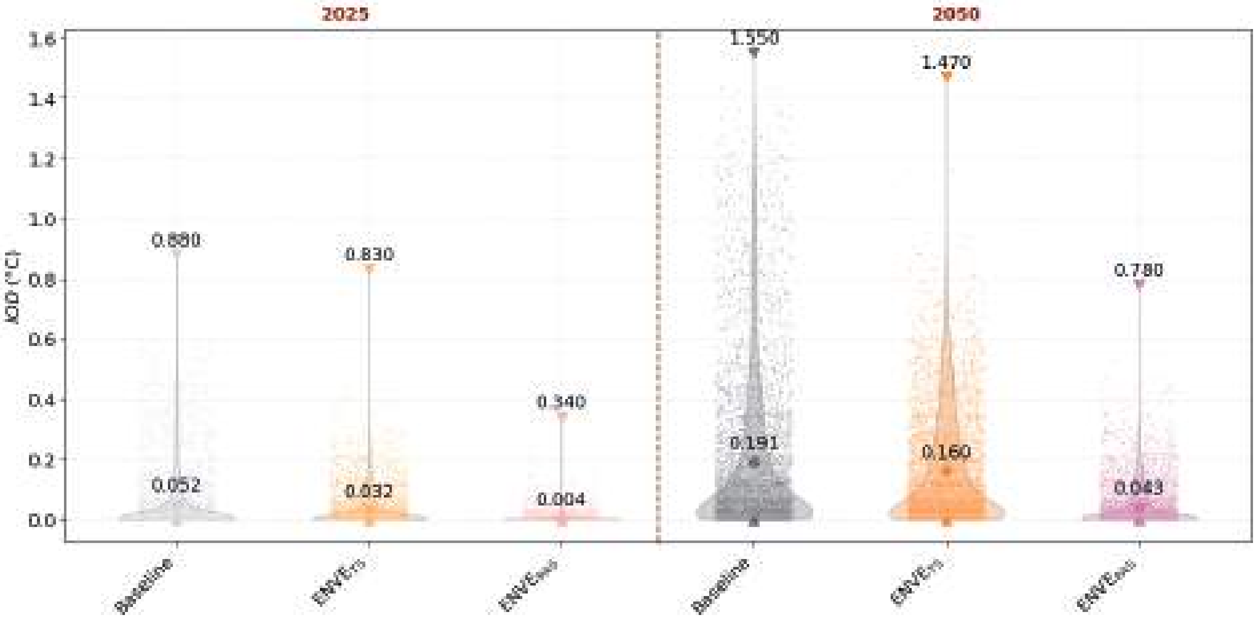


The scenarios employing fossil-fuel-based heating (Baseline, $ENVE_{TS}$, and $ENVE_{PAS}$), which exclude the substantial heating load from the total electricity demand, exhibit deceptively high self-sufficiency, approximately 80–81% in 2025 and 75–78% in 2050. These values are not indicative of high self sufficiency, and must be viewed with caution as buildings' heating demand is met not by electricity but by natural gas.

The scenarios involving HP take into account heating energy consumption as part of the total electricity energy load, showing the self-sufficiency index as 52% in 2025 and 58% in 2050 for the HP + PV scenario. The index rises from 64-69% to 67-72% for the $ENVE_{TS}$ + HP + PV and $ENVE_{PAS}$ + HP + PV scenarios from 2025 to 2050. This positive trend in which the combined scenarios of envelope renovations and HP demonstrate the highest achievable self-sufficiency in fully-electrified systems.

6.2. The Results of Thermal Comfort Indicators and Energy Renovation Strategies

Indoor Overheating Degree

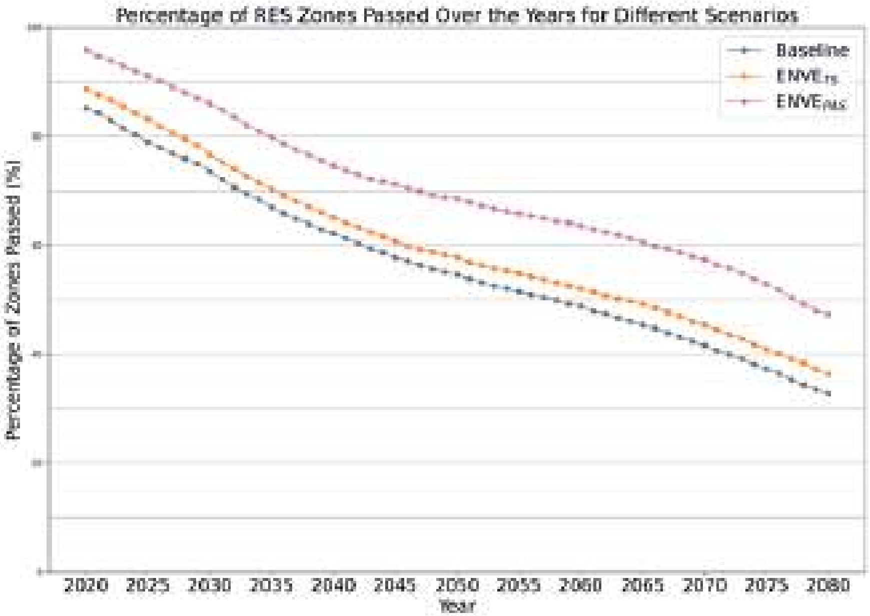


The analyses of Indoor Overheating Degree (IOD), focuses on the TS and Passivhaus scenarios, as HP and PV integration does not directly influence overheating risk. From 2025 to 2050, the Baseline scenario shows a consistent upward trend, with the mean IOD increasing nearly fivefold, highlighting the growing threat of overheating under future climates.

A closer look at the renovation scenarios shows that in 2025, TS insulation almost halves the Baseline mean, while Passivhaus insulation reduces it to nearly zero. By 2050, however, TS insulation remains limited, lowering the mean IOD by only about 16%, whereas Passivhaus insulation achieves a 77% reduction.

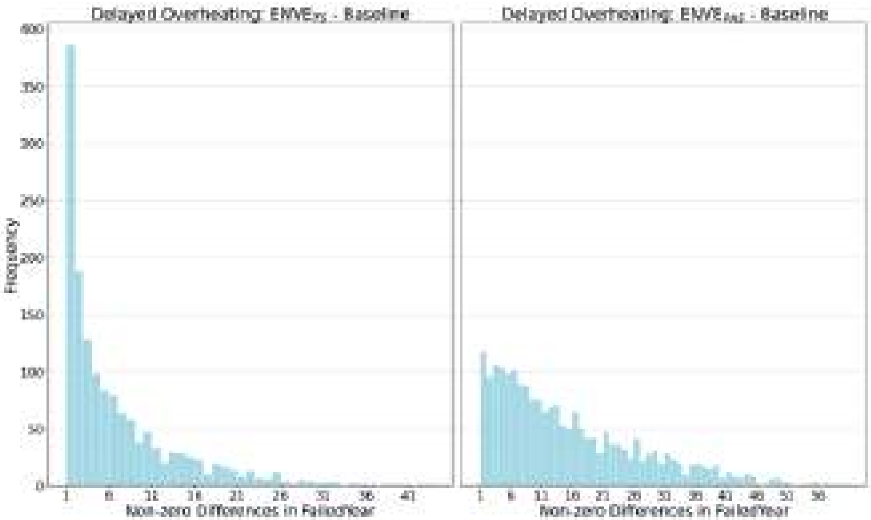
Overall, while the Baseline reveals a strong trend toward increasing overheating, the insulation strategies demonstrate clear mitigation potential. Passivhaus envelope insulation performs best across both timeframes, keeping overheating close to lower levels in 2025 and maintaining its effectiveness by 2050.

Percentage of Zones Passing The CIBSE Overheating Criteria



The risk of overheating can be assessed using the **CIBSE (The Chartered Institution of Building Services Engineers)** overheating risk test, which classifies a living space as failed if the number of overheated hours exceeds 3% of the occupied hours. In the baseline scenario, about **85%** of non-air-conditioned living spaces fulfil the CIBSE overheating criteria in 2020, but by 2050 and 2080, this percentage is 54% and 32% respectively. Energy renovation scenarios **help to delay the year** in which the overheating risk materialises in some cases.

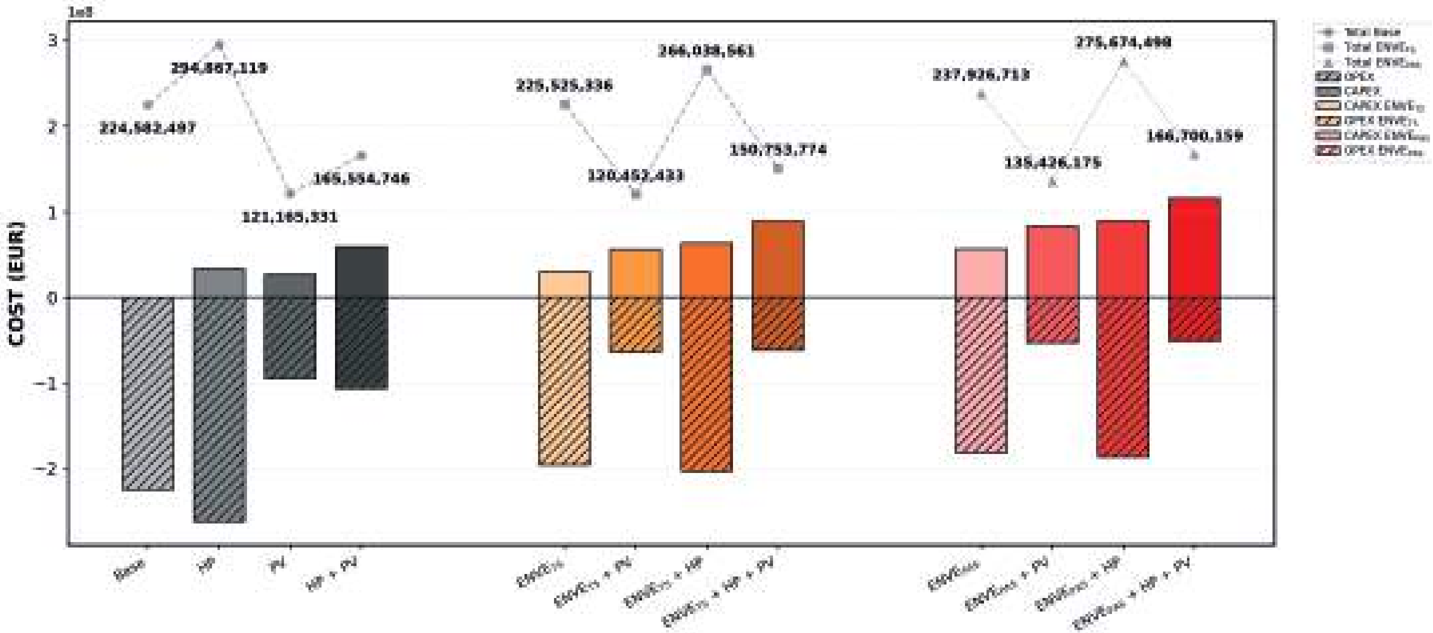
Delayed Overheating



The distribution of the number of years by which overheating is delayed as a result of TS (left) and Passivhaus (right) can be seen to the left. It shows the shift towards long delay periods when renovated with TS standards and even longer delay periods with the Passivhaus standards. TS and Passivhaus strategies reveal that insulation quality determines resilience, demanding joint assessment of energy and indoor thermal performance.

6.3. The Results of Economic Indicators and Energy Renovation Strategies

CAPEX, OPEX and Total Costs (EUR)



ROI shows the inverse relationship between Capital Expenditures (CAPEX) and Operational Expenditures (OPEX). In the baseline scenario, costs are only due to OPEX, around €225 million. PV scenario reaching one of the lowest total costs, around €121 million. However, adding only the HP systems drives both CAPEX and OPEX upward, with total expenditures nearly €295 million, as electricity demand increases. The most effective strategy is the $ENVE_{TS}$ + PV scenario, the lowest total cost, approximately €120 million. Although Passivhaus renovations offer long-term emission reductions more effectively than the TS insulations, they also require a higher initial investment.

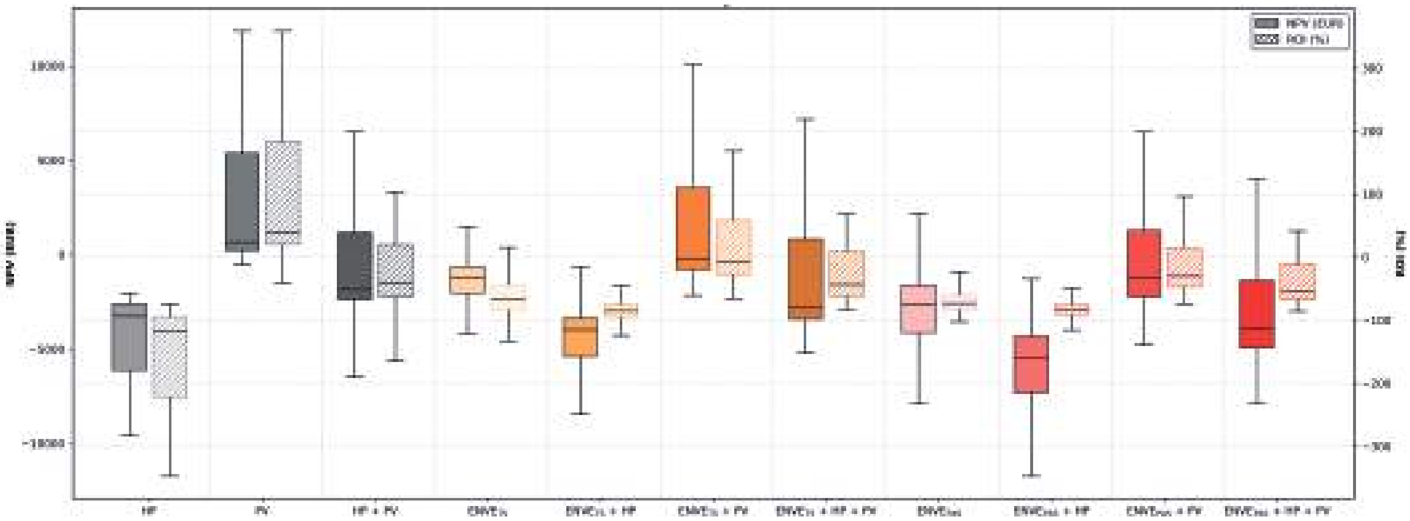
Key Takeaways

•Under the tiered electricity tariffs, electrification of heating pushes total consumption above threshold limits and significantly increases electricity bills, making heat pump integration economically unfeasible.

•Super-insulated buildings demand almost twice the initial investment compared to standard insulation. This scenario becomes economically and environmentally justified only when the objective is to reach NZEB performance levels.

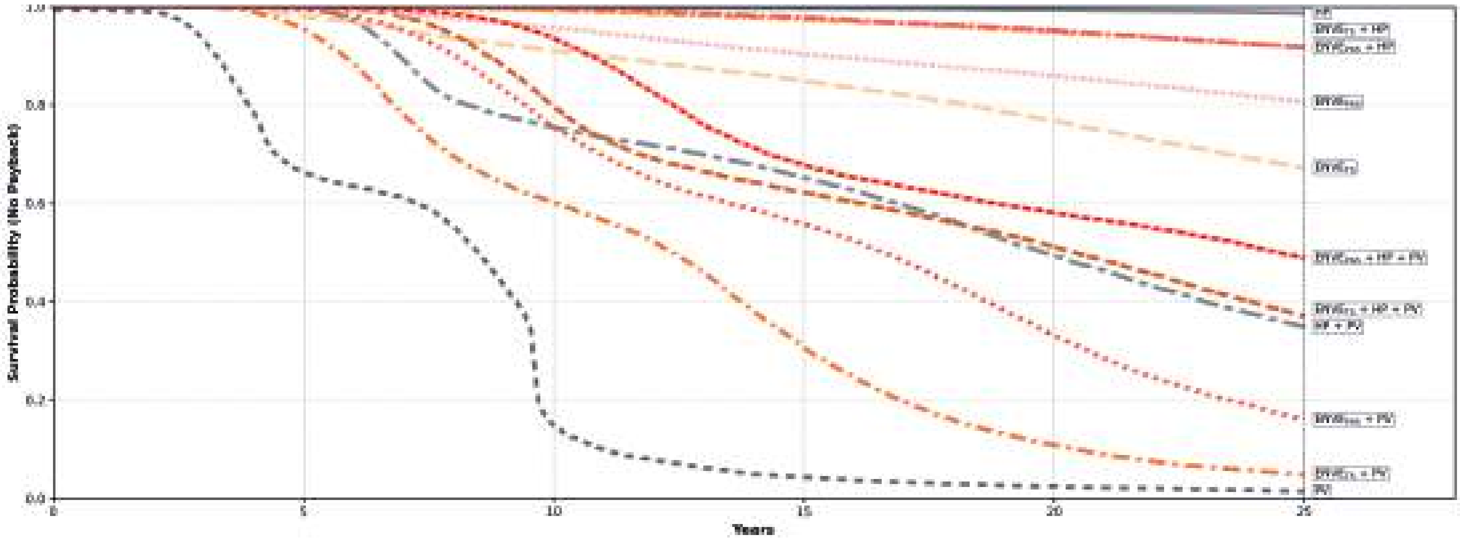
•Electrification of heating systems becomes economically feasible only in PV combined scenarios, due to the electricity tariff implementation.

NPV and ROI by Scenario



The Net Present Value (NPV) and Return on Investment (ROI) distributions show that combining PV with Baseline and $ENVE_{TS}$ scenarios are economically advantageous, showing median NPVs at or above zero and median ROIs between 20–60%. In contrast, HP alone and most Passivhaus scenarios remain financially unattractive: NPVs are predominantly negative and median ROIs are below zero, even when PV or HP is added. All scenarios with PV systems, particularly the $ENVE_{TS}$+ PV scenario, offer the most compelling economic returns. Envelope renovations combined with HP, without PV integration ,$ENVE_{TS}$ + HP and $ENVE_{PAS}$+ HP, exhibit the weakest performance.

Payback Survival Curves by Scenario



In the baseline scenario with PV, half of the buildings achieve payback in under 10 years. Adding PV to HP and TS envelope renovations similarly reduces the mean payback time to around 12–15 years. In contrast, scenarios such as $ENVE_{TS}$ + HP renovations show a more gradual decline in payback probability, with the mean payback not reached within 25 years. Passivhaus scenarios, even with HP, demonstrate a slower financial return, with over 80% of projects not recovering their initial investment within two decades. The Discounted Payback Period allows users to balance long-term sustainability goals with realistic financial timelines.

Key Takeaways

•If the initial capital is limited, PV system installation is the most feasible approach, followed by adding TS insulation. This combination provides faster financial returns while maintaining acceptable energy performance.

•To achieve financial return over a 25-year period, scenarios that include PV systems should be prioritized.

Marginal Abatement Cost Curve by Scenario

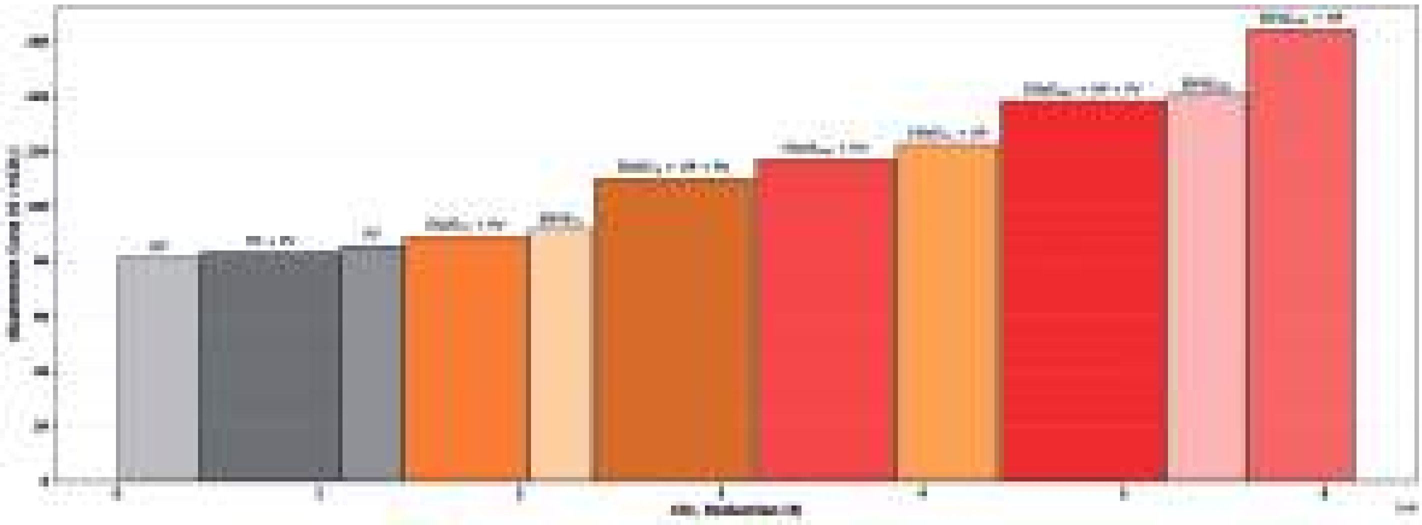
The Abatement Cost shows the cost-efficiency of carbon reduction. The graph illustrates a clear trade-off between reduction potential and cost per ton of CO_2 abated. On the x-axis, the graph shows the CO_2 reduction potential, while the y-axis indicates the abatement cost. According to the results, PV offers the lowest abatement cost at approximately €82 per ton CO_2, with a total reduction of around 0.5 million tons. The integration of PV generally increases the unit cost slightly but significantly increases the total avoided emissions. For instance, adding PV to HP raises abatement costs marginally but extends the total reduction to over 1.0 million tons. Similarly, $ENVE_{TS}$ + PV balances moderate costs with larger reduction potential. While $ENVE_{TS}$ insulation alone is more expensive than HP or PV scenarios, combined scenarios like $ENVE_{TS}$ + HP + PV yield higher emission savings, at rising costs. Passivhaus-level insulation stands out as the most expensive one, with $ENVE_{PAS}$ + HP reaching nearly €182 per ton CO_2, about twice as costly as the HP scenario.

Key Takeaways

•Although PV integration offers the most cost-effective outcome in terms of emission abatement, its emission reduction remains lower than other combined scenarios.

•If insulation scenarios are not preferred, a combined HP + PV scenario may represent a cost-effective alternative in terms of abatement cost.

•When the main objective is emission reduction, combined scenarios deliver the highest overall performance. Among these, the Passivhaus configuration with HP and PV systems performs slightly better, owing to its higher insulation capacity.



7
E-SCOOTER

7.1. Data Analysis

Identifying optimal charging locations for e-scooters is essential to ensure efficient energy use and support the broader transition toward sustainable urban mobility. Strategic placement of charging stations not only reduces operational downtime and grid dependency but also enables the integration of renewable energy sources for cleaner charging infrastructure. In this context, the assessment of rooftop solar potential plays a key role in identifying areas where on-site renewable generation can directly support e-scooter charging demand. The spatial correlation between high solar potential and mobility demand enhances the feasibility of decentralized, low-carbon charging hubs. To achieve this, a data-driven approach is applied by combining mobility demand patterns with renewable energy potential using two main datasets.

The first dataset consists of detailed e-scooter trip records from 2022. The end points of these trips indicate where people actually park, which serves as a real-world indicator of mobility demand. A spatial density analysis of these parking points reveals clear "hotspots," showing neighborhoods with the highest concentration of e-scooter usage. These hotspots highlight where charging stations would be most useful.

Parking Locations of E-scooters in Kadıköy

Building Electricity Consumption Map of Kadıköy

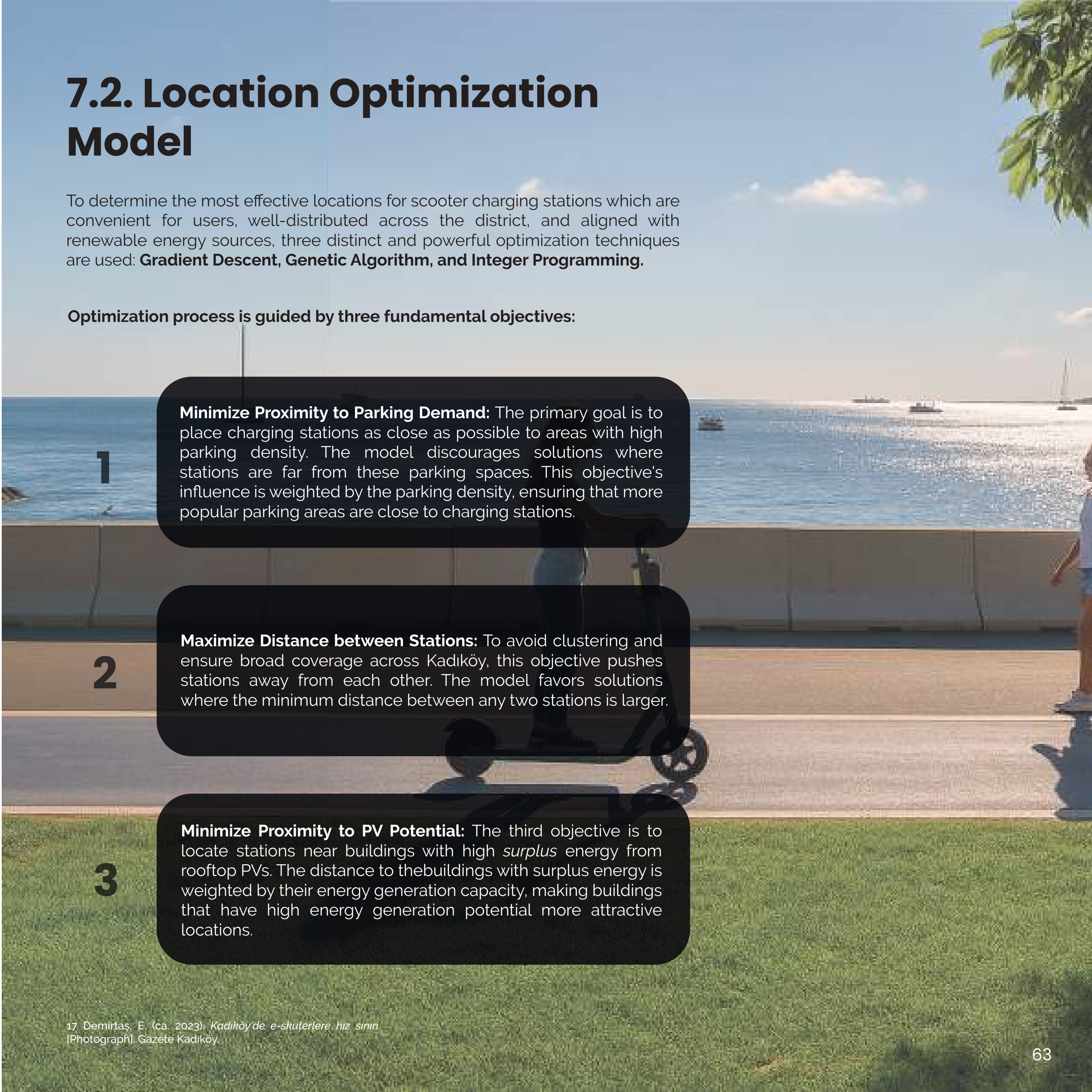
Rooftop Photovoltaic Energy Generation Map of Kadıköy

Surplas Electrical Energy Map of Kadıköy

The second dataset is based on Geographic Information Systems (GIS) records and 3D building models from the Kadıköy Municipality. This data provided critical information such as building locations, footprint areas, and usage types (residential or office). This information was used to estimate two key metrics for each building: its **annual electricity consumption** and its potential for **rooftop photovoltaic energy generation.** By subtracting the building's consumption from its potential generation, we calculated a third metric: **the potential surplus electrical energy** for each building.

These energy calculations are visualized on the map of Kadıköy. An examination of the building electricity consumption map reveals that consumption is concentrated in areas with commercial axes and densely populated residential zones. The solar energy potential map indicates that buildings with large and unobstructed roof areas possess high generation potential. The surplus electrical energy map, which presents the most significant findings, clearly identifies the areas with energy surpluses and deficits within the district. This suggests that clusters of buildings with an energy surplus could serve as potential hubs for future local energy communities.

Numerically, the mean potential for solar generation (29,352.8 kWh) was calculated to be slightly higher than the mean building electricity consumption (25,519.5 kWh). In terms of net energy, it was observed that 32% of the buildings had no potential for a surplus. The remaining buildings, however, had a mean surplus of 16,782 kWh. In the optimization algorithms, these buildings with an energy surplus are considered to have high PV potential.



7.2. Location Optimization Model

To determine the most effective locations for scooter charging stations which are convenient for users, well-distributed across the district, and aligned with renewable energy sources, three distinct and powerful optimization techniques are used: **Gradient Descent, Genetic Algorithm, and Integer Programming.**

Optimization process is guided by three fundamental objectives:

1. **Minimize Proximity to Parking Demand:** The primary goal is to place charging stations as close as possible to areas with high parking density. The model discourages solutions where stations are far from these parking spaces. This objective's influence is weighted by the parking density, ensuring that more popular parking areas are close to charging stations.

2. **Maximize Distance between Stations:** To avoid clustering and ensure broad coverage across Kadıköy, this objective pushes stations away from each other. The model favors solutions where the minimum distance between any two stations is larger.

3. **Minimize Proximity to PV Potential:** The third objective is to locate stations near buildings with high *surplus* energy from rooftop PVs. The distance to thebuildings with surplus energy is weighted by their energy generation capacity, making buildings that have high energy generation potential more attractive locations.

17 Demirtaş, E. (ca. 2023). *Kadıköy'de e-skuterlere hız sınırı* [Photograph]. Gazete Kadıköy.

Optimization Techniques

1. Gradient Descent

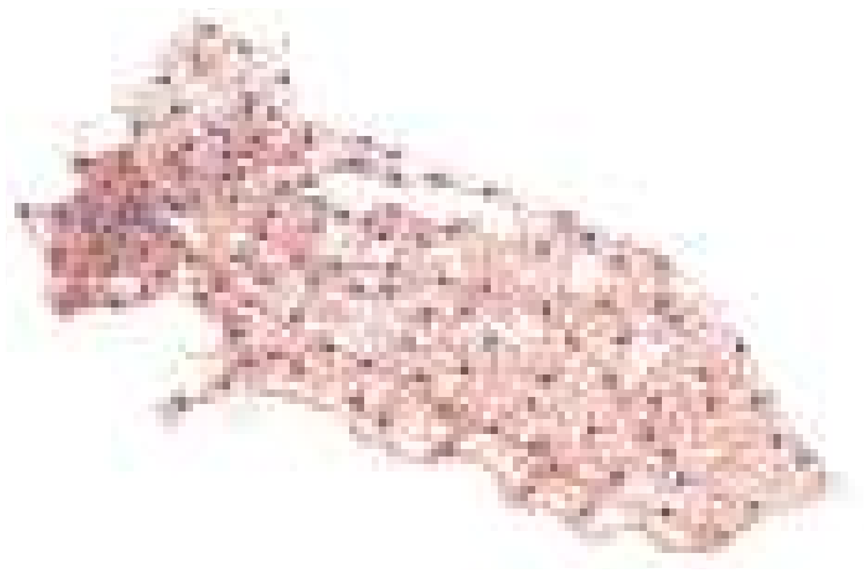
It is an iterative algorithm for continuous optimization. The process begins with a set of station coordinates, which are then progressively refined. In each step, the weighted sum of the three objectives is calculated by the algorithm; this function is called "loss function", and to minimize the loss most effectively the station coordinates are adjusted. This cycle is repeated hundreds of times, allowing the station layout to converge toward an optimal configuration that balances all competing goals.

2. Genetic Algorithm

The principles of natural selection are the inspiration of this method. It starts with creating a large "population" of diverse potential solutions of station layouts. Each solution is then evaluated with a "fitness function" that scores how well it satisfies the three objectives. Through crossover and mutation each solutions' best traits are combined to create a new, superior generation Then, the "fittest" solutions are selected to "reproduce". This process continues until a highly optimized population is achieved.

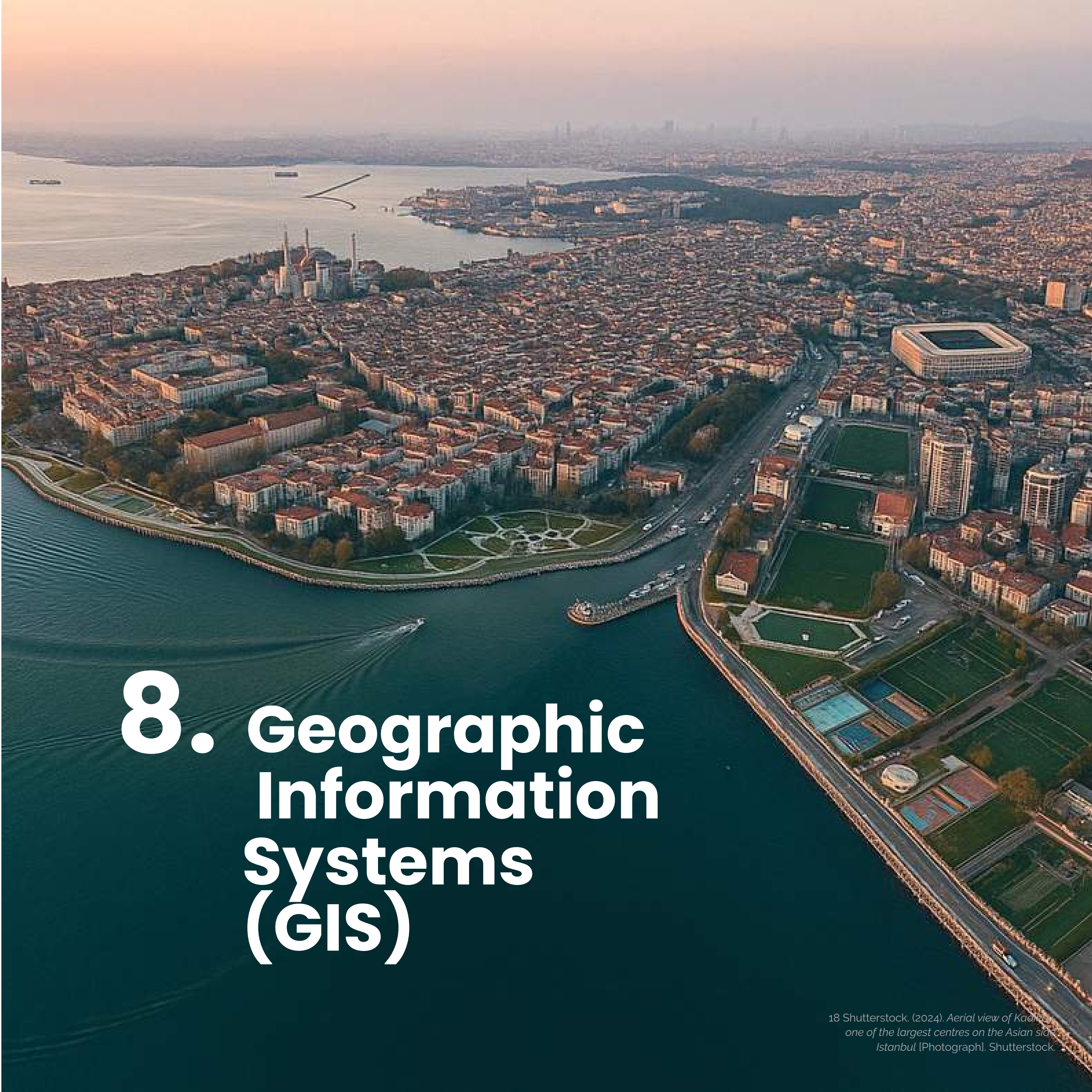
3. Integer Programming

Integer Programming provides a more formal and precise approach. For this method, firstly a large set of discrete candidate locations for the stations is defined. The problem is then formulated as a system of linear equations where the model must make a binary (Yes/No) decision for each candidate spot. Its goal is to select the combination of locations that satisfies the three objectives well, while adhering to constraints such as placing a specific total number of stations.



Optimal Charging Station Locations in Kadıköy

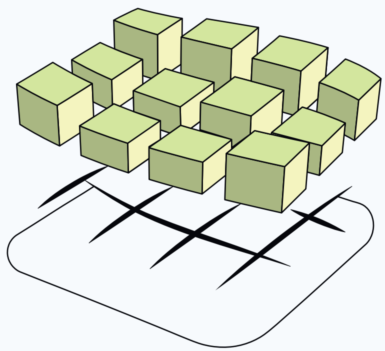
Finally, the figure illustrates the optimal placement of 150 e-scooter charging stations as determined by the Gradient Descent optimization model. The blue dots represent the optimized station locations, distributed across the district to balance the three core objectives which are explained above. This map provides a tangible visualization of the model's solution, highlighting strategic points for sustainable charging infrastructure deployment in Kadıköy.



8. Geographic Information Systems (GIS)

18 Shutterstock. (2024). *Aerial view of Kadikoy, one of the largest centres on the Asian side, Istanbul* [Photograph]. Shutterstock.

8.1. Geographic Information Systems (GIS)



Geographic Information Systems (GIS) offer significant advantages for urban-scale analysis, particularly through the web-based visualization of linked data layers of diverse datasets that incorporate both geometric and semantic data with precise location specificity. The integrated geospatial capabilities of GIS, which facilitate the interactive exploration of KPIs and provide immediate visual feedback, position GIS as a great medium for displaying UBEM simulation data. By the use of geospatial data, GIS enables a clearer and more accessible understanding of energy efficiency and urban resilience for a wide range of stakeholders. Crucially, GIS' ability to integrate with other datasets and GIS-based applications provides decision-makers with comprehensive insights such as, grid networks, transportation routes, population density, and demographics.

GIS capabilities are further enhanced by features that allow the model to be **seamlessly integrated with** mixed **urban data**. A GIS-based interface, complete with linked data layers, ensures the reliable accessibility of both analytical and descriptive data. Within this framework, the visualization of KPIs, which covers both the individual and cumulative energy behaviors of buildings, serves as a central component for data-driven urban-scale decision-making. **The key benefits and features of GIS are :**

3D Model Interaction: Users can browse through the 3D city model, zoom in and out, and click on buildings, zones, or roofs to access semantic data and related KPIs.

Standardization and Integration: As the system is built on a GIS infrastructure, it follows common standards and can be integrated with other GIS-based datasets and applications.

Expressive Data Visualization: Provides a clear, meaningful, and visually strong representation of data.

Interactive Exploration: Enables filtering, layer control, and scenario comparison in an integrated way.

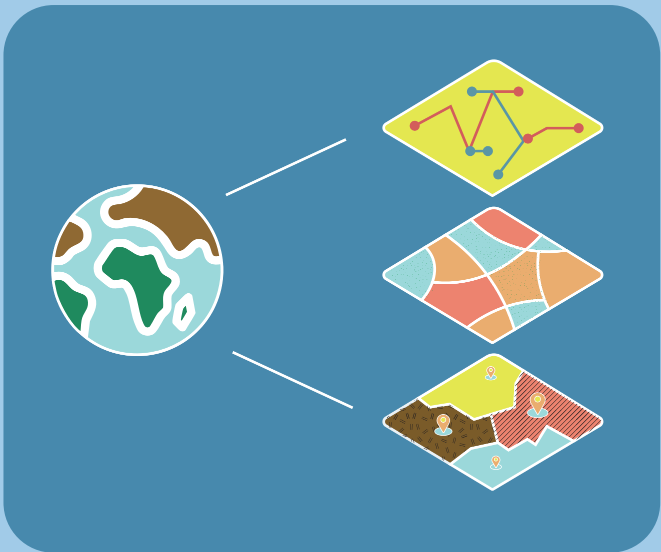
Accessibility for Stakeholders: Combines geometry and semantic data in a single environment, making the results understandable for all stakeholders.

Geospatial Data

Geospatial data consist of the following components:

•Coordinates of model entities
•Geometric shapes such as buildings, roofs, and zones
•Data attributes, such as building-related information and calculated KPIs

Geospatial data for each individual thermal zone, building, and roof enables the model to be displayed in its exact location with 3D geometry and linked attributes, forming the framework of the GIS-based decision-making tool.



19 Kadıköy'ün tarih kokan sokakları. (2023). [Photograph]. Pinterest. [AI genereated image].

8.2. The Decision Making Tool for Decarbonization

The tool, developed in collaboration between METU and ODTÜ-GÜNAM, is designed to support urban-scale decision-making and accelerate energy renovation efforts, especially in response to the long-term impacts of climate change. The tool's main goal is to evaluate different renovation scenarios by turning complex, urban-scale simulation data and KPIs into clear, visual insights that can help through the decision-making process.

The tool is composed of three key enabling technologies including Urban Building Energy Modeling (UBEM), Artificial Intelligence (AI), and Geographic Information Systems (GIS). The tool can help evaluate various renovation scenarios that implement passive systems, active systems, and renewable energy system integration. Finally, the tool supports the calculation and visualization of KPIs that a broad range of decision makers can understand.

8.3. Overview of the Dashboards

Brief Overview

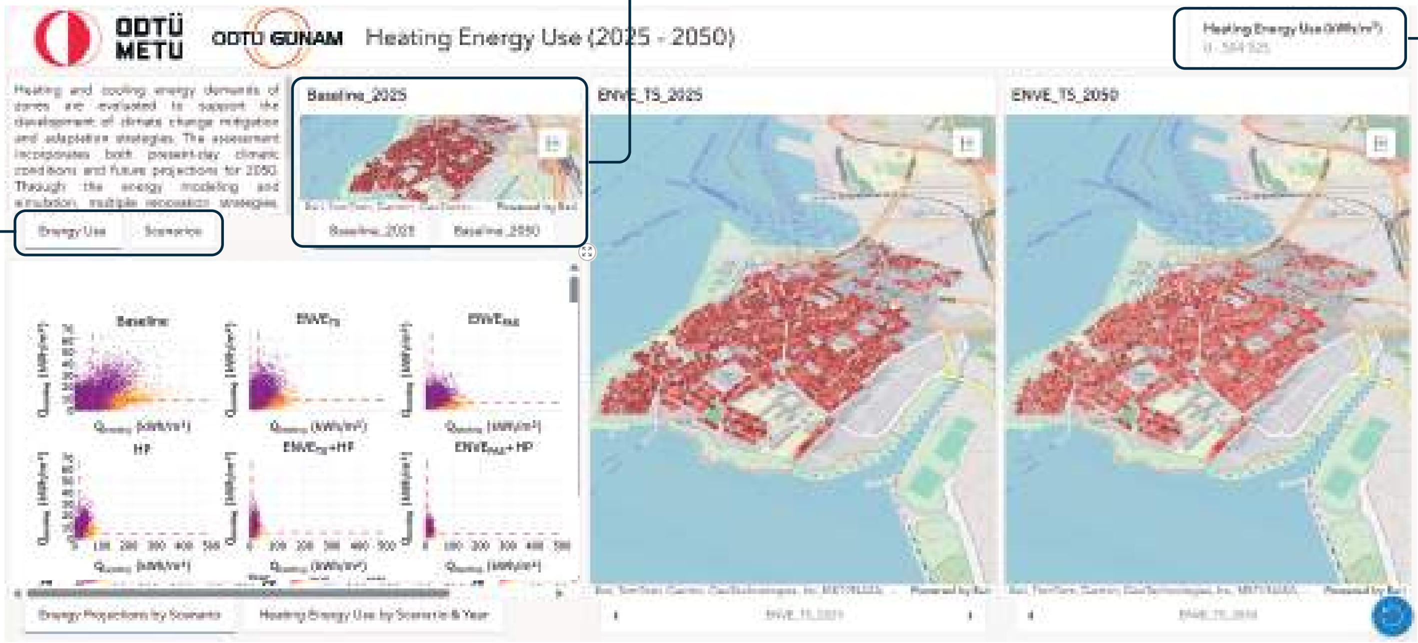
Detailed information regarding the selected KPIs is provided on the dashboard, along with all energy renovation scenarios.

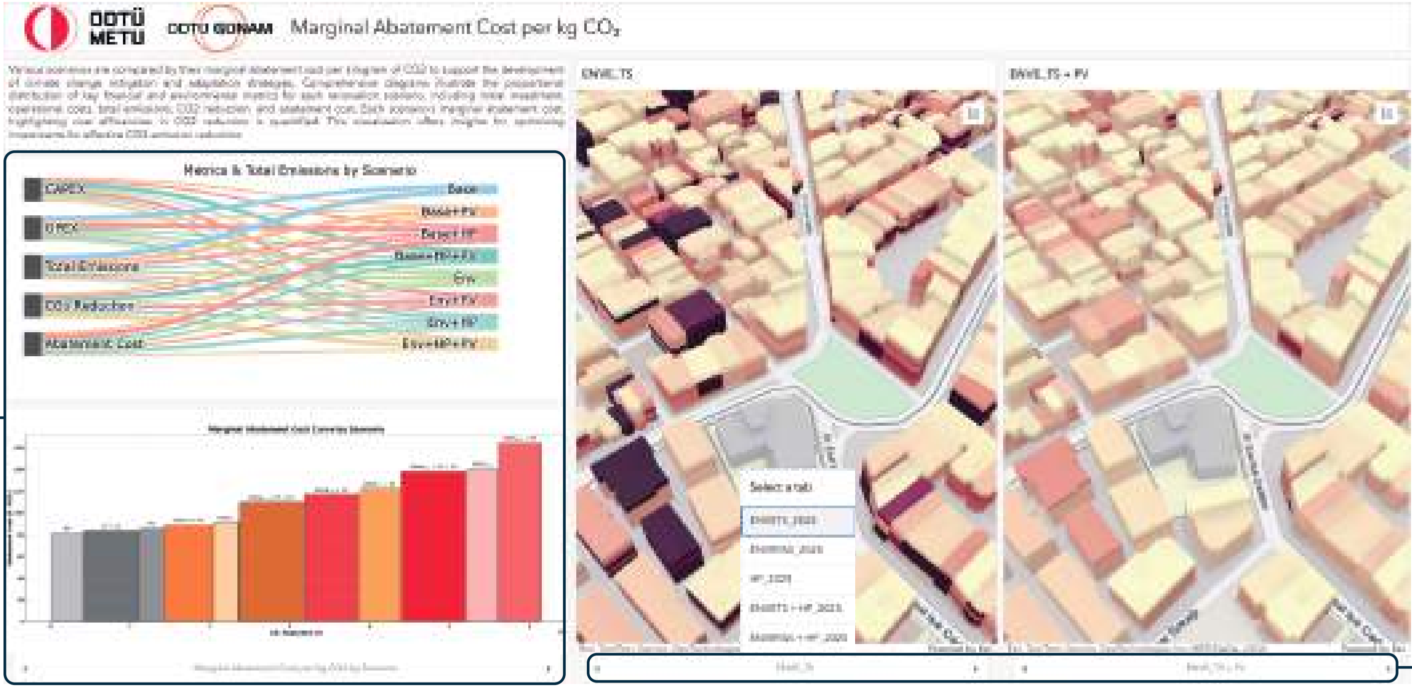
Comparison of the 2025 & 2050 Results

Users can compare the results for the years 2025 and 2050. By zooming in on the buildings, a side-by-side analysis becomes possible.

KPI Filters

Applying filters allows users to narrow the result range, enabling a focused analysis of specific building groups based on defined criteria.





Graph Visualization

The provided graphs enable users to explore the KPI results and evaluate various energy renovation scenarios.

Scenario Selection

Synchronous, side-by-side comparison of the renovation scenarios is enabled within the dashboard through 3D visualizations of the UBEM simulation results for selected KPIs for the years 2025 and 2050, allowing users to closely examine differences between the years.

Relevance to Cities:

How does this tool support the three pillars of UP2030?

Carbon Neutrality

•The tool contributes to carbon neutrality by providing data-driven decision support, primarily through scenario analysis. Supporting municipal authorities to identify and define decarbonization pathways that are aligned with the European Green Deal and the UP2030 objectives.

•The tool quantifies environmental impact with KPIs, showing these indicators such as total CO_2 emissions and reductions for each energy renovation scenario.

Just Transition

•The tool supports clean energy transition by integrating affordability indicators and cost-benefit analyses, ensuring that renovation measures remain accessible and equitable across different income groups.

•Provides filtering and KPI-based analysis to prioritize different renovation needs, while 3D displays enhance traceability and spatial understanding of the renovation scenarios.

Resilience

•Evaluates the performance of buildings under changing climate change conditions, allowing cities to foresee risks and increase their capacity for adaptation through data-driven analysis.

•Encourages clean energy use through rooftop PV integration, supporting energy security and reducing dependence on carbon-intensive energy sources.

•Provides metrics such as indoor overheating degree (IOD) to show potential risks associated with rising temperatures and to assess how energy renovations impact thermal comfort.

20 Gazete Kadıköy. (2022). *Arabulucu çözüm olur mu?* [Photograph].

Benefits and Synergies:

What are the key benefits for cities and how does it create synergies with other project pillars?

Digitisation

Combining advanced digital tools with GIS data to enable decision-makers exploring various energy renovation scenarios which makes the planning process more inclusive and collaborative.

Spatial Justice

Helping to identify areas with the highest energy consumption and enabling targeted action where it's needed most.

Sustainable Urban Energy Planning UBEM

shows how renovation strategies and solar-powered mobility can support cities in transitioning toward a greener, low-carbon future.

8.4. Challenges

1. Computational Demand

The computational demands on client-side hardware and limitations in server-side capacity lead to operational latency. This presents a significant challenge to both system performance and data visualization due to computational cost.

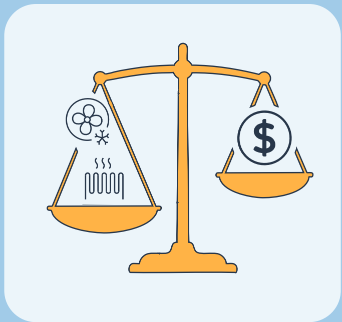
2. Data-related Challenges

Dataset development is complicated by the use of multiple, heterogeneous sources lacking a standard and well-structured format. This required data augmentation to ensure a statistically robust and representative dataset.

3. Expertise and Modeling Cost

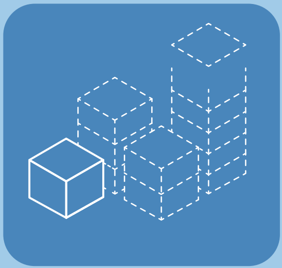
UBEM modeling and simulations are computationally intensive and require specialized expertise. This means that the tool's use in each new city requires significant effort for data collection and modeling, as well as new dataset development.

4. User-Side Priorities



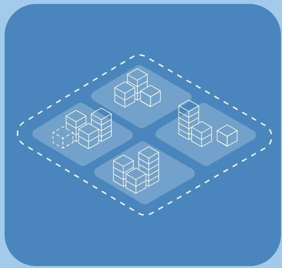
The tool currently allows the exploration of only the key renovation scenarios. The tool's flexibility and broad-scale applicability can be enhanced by facilitating users' real-time scenario definition.

8.5. Main Takeaways



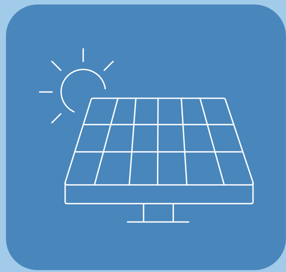
1

Large-scale renovation is essential as existing building stock is often energy-inefficient and constructed under outdated standards, making isolated investments insufficient for meaningful improvements.



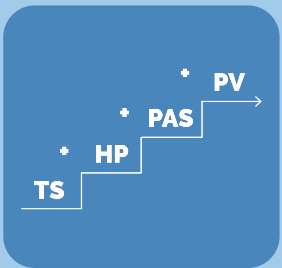
2

A city-scale, scenario-based approach is critical for balancing environmental, economic, and comfort key performance indicators through the comprehensive comparison of all viable options at both the building and neighborhood levels.



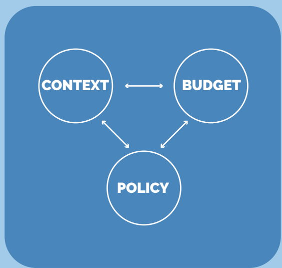
3

Making photovoltaics a standard practice consistently lowers total costs and reduces CO2 emissions. Improving the self sustainability of the buildings and the neighborhood.



4

Implementing a stepwise renovation pathway, beginning with insulation and progressing to the integration of heat pumps and photovoltaics, is an effective strategy to maximize energy reductions and occupant comfort.



5

There is not one single "BEST" solution: optimal choices depend on a balance of competing key performance indicators, with decisions shaped by contextual factors, budgetary constraints, and environmental policies.

6

User-friendly interfaces can help a wide range of stakeholders better understand complex data and its implications by presenting information clearly and intuitively, these interfaces allow users to explore the data in great detail and grasp its significance.

9
PV-INTEGRATED URBAN FURNITURE

9.1. Objectives of Urban Furniture

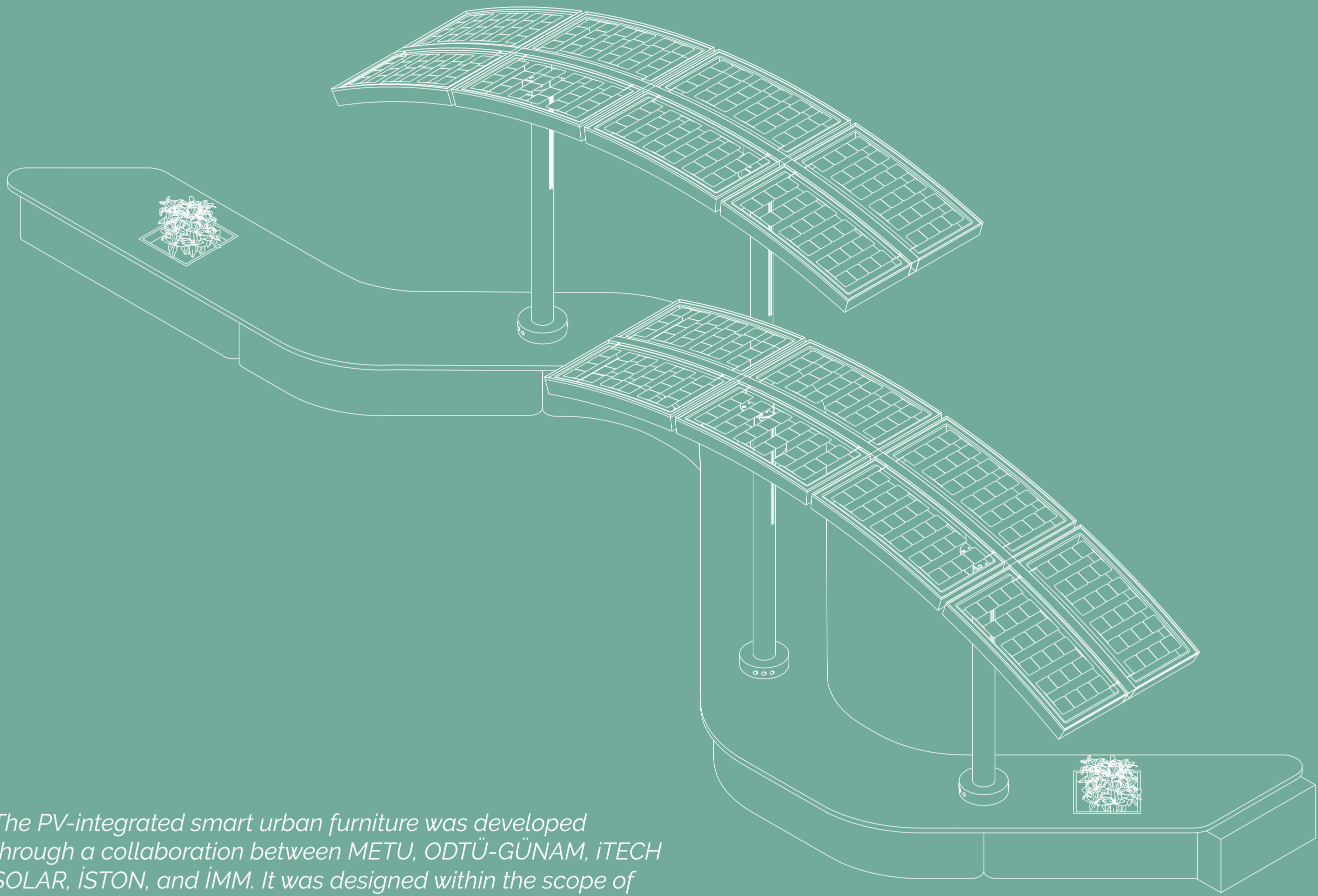
Public outdoor spaces play a key role in raising **neighborhood** awareness and forms **collective responsibility** toward sustainable energy use. The urban furniture design integrates PV technology with public space to address challenges related to **energy, sustainability**, and **social engagement**. The proposed design offers solutions for climate resilience goals and smart city strategies with renewable energy production and public interaction.

By generating approximately 1.6 MWh of solar electricity per unit annually, the PV-integrated urban furniture contributes to urban sustainability by decreasing the need for fossil-fuel usage. Its modular configuration enhances the functionality and accessibility of public spaces by incorporating durable seating with shading, high-efficiency LED illumination, and USB charging stations to support extended use. The public furniture also aims to raise awareness of green energy through the visible and functional integration of photovoltaic technologies.

9.2. Design and Technical Overview

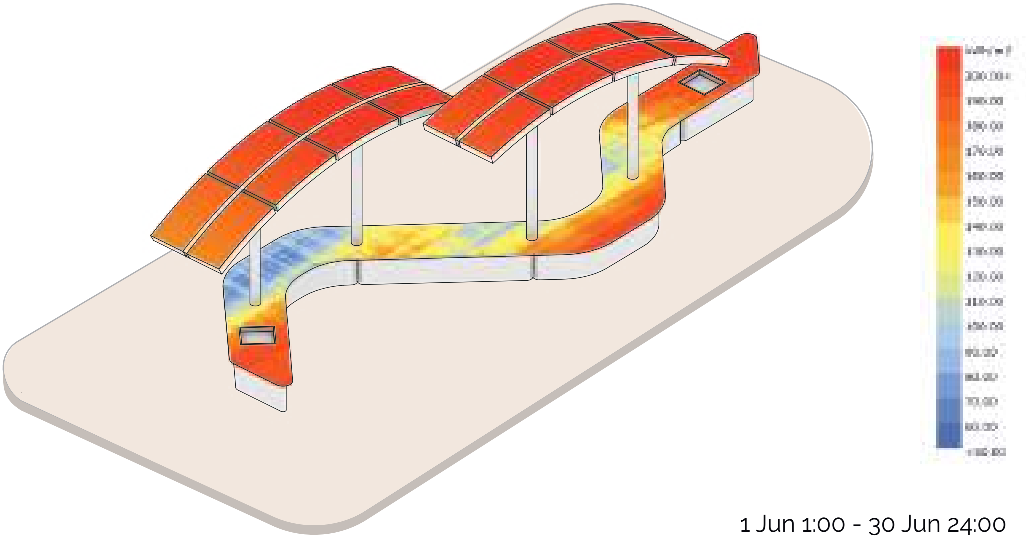
Design Features And Applicability

The furniture incorporates a clean, modular design suitable for all users.
Materials include corrosion-resistant steel, laminated safety glass, wood panels and a 3D concrete bank designed for outdoor use with a **long lifespan** and **minimal maintenance**. The 3D-printed concrete structure is supported by steel elements carrying a PV panel roof made of **8 identical, single-curved panels.**

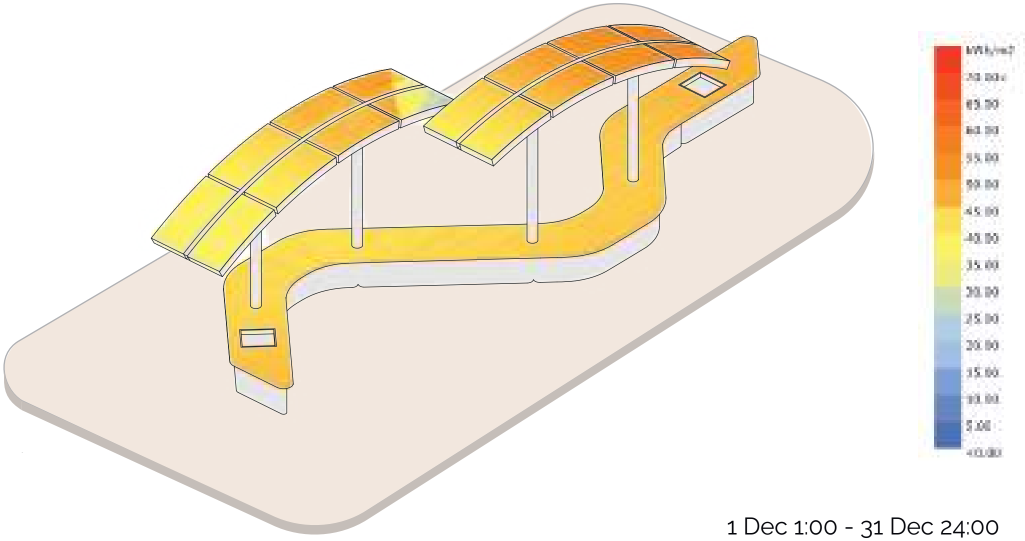


The PV-integrated smart urban furniture was developed through a collaboration between METU, ODTÜ-GÜNAM, iTECH SOLAR, İSTON, and İMM. It was designed within the scope of Yasin Kantaş's master's thesis.

Total Radiation Analysis



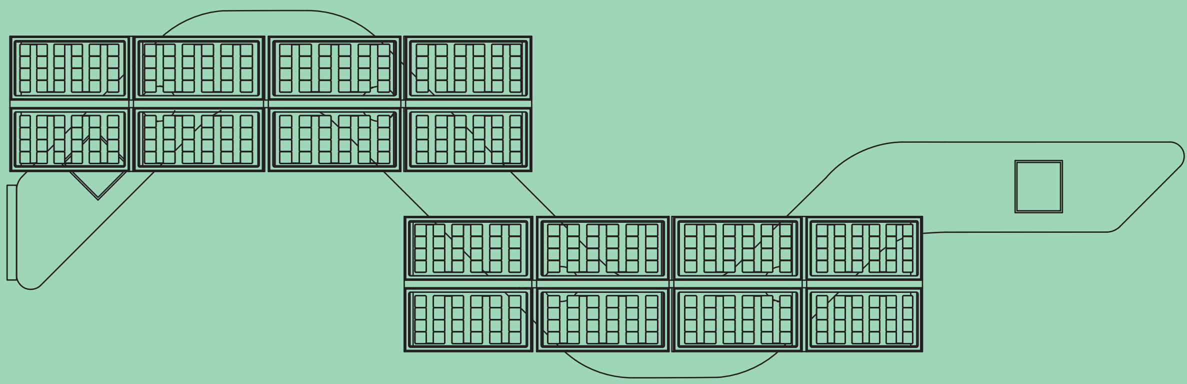
1 Jun 1:00 - 30 Jun 24:00



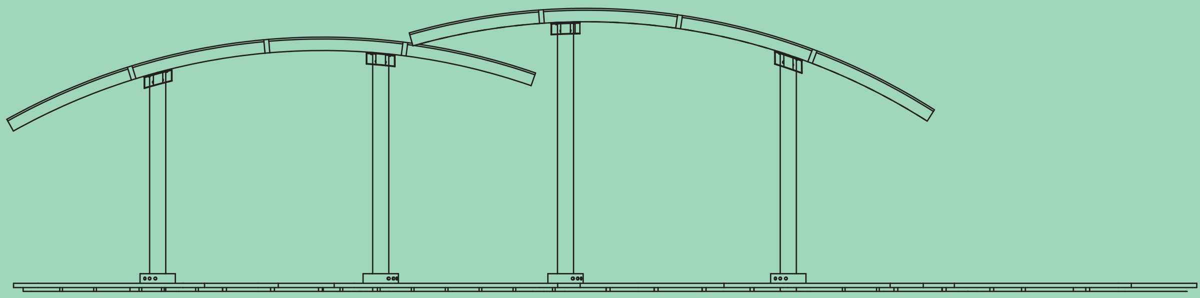
1 Dec 1:00 - 31 Dec 24:00

The analysis reveals clear seasonal differences in solar exposure across the urban furniture, with high radiation concentrated on the panels during summer and a more uniform distribution observed in winter due to lower solar angles. The analysis are conducted as part of Yasin Kantaş's master's thesis.

9.3. PV Integration



TOP VIEW



FRONT ELEVATION

- **PV Panel Dimensions:** 60x135 cm, 80W each, frameless and semi-transparent
- **Total Area:** 14 m^2
- **Total Power:** 1280W
- **Expected Output:** Approx. 1,600 kWh/year based on Istanbul conditions
- **Orientation:** South-facing, 25° tilt angle

9.4. Manufacturing Stages

Smart Components

The furniture is equipped with:

•USB charging ports powered by PV

•LED lighting

•MPPT tracking, temperature, and irradiance sensors

•IoT connectivity using low-energy transmission (LoRa or NB-IoT)

Sustainability and Environmental Impact

The urban furniture helps reduce carbon emissions, eliminates the need for fossil-based energy in public spaces, and fosters clean energy literacy. The design emphasizes:

•Use of recycled and recyclable materials

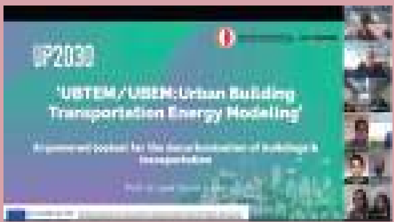
•Durability exceeding 25 years

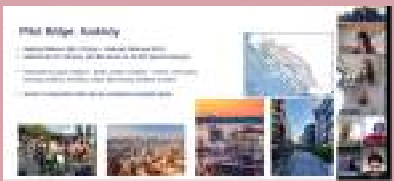
•Estimated annual CO_2 savings of 700 kg per unit

10. WORKSHOPS

10.1. Needs and Barriers Workshop

This workshop brought up the opinions of stakeholders to support Istanbul's transition toward carbon neutrality at the neighborhood level. Bringing together local authorities, citizens, experts, and businesses, the workshop identified critical barriers, such as regulatory hurdles, financial constraints, and infrastructure limitations, as well as opportunities, including urban transformation initiatives, public awareness, and international collaboration. It also outlined realistic solutions, like enhanced legislation, financial incentives, micro-mobility infrastructure, and renewable energy integration. These findings form the basis for context-sensitive strategies that foster collaboration, innovation, and sustained progress in reducing urban carbon emissions.





10.2. Vision Workshop

Following initial evaluations by METU, ODTÜ-GÜNAM, and İMM, Kadıköy was selected as the UP2030 pilot area due to its rich data availability and its relevance to vulnerable groups. The project's foundation was strengthened through a vision workshop and survey, which revealed low public awareness of climate goals but a strong interest in solar energy adoption. Inadequate visibility of stakeholders and the lack of community engagement brought up several recommendations, such as encouraging awareness campaigns, promoting renewable technologies, and clarifying stakeholder roles. With these insights, Kadıköy's pathway toward sustainability was shaped through integrated spatial analysis, participatory visioning, and adaptive planning. Stakeholders co-developed the carbon neutrality vision, emphasizing resilience, inclusivity, and transition to clean energy. These steps ensured that Kadıköy's transformation is inclusive, data-driven, and technically sound, paving the way for a resilient and carbon-neutral urban future.

10.3. Action Workshop

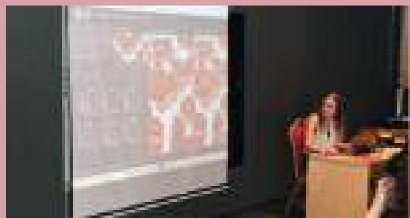
The workshop took place at the Gazhane Museum in Istanbul. Stakeholders and citizens from Kadıköy are gathered to develop strategies supporting a climate-neutral future. Discussions focused on energy efficiency in buildings, short-term climate risks, and ways to raise public awareness for renewable energy and PV-integrated urban furniture. Rooftop PV integration, smart lighting, community incentives, educational campaigns, and interactive tools were the key proposals to promote behavioral change. By grounding pilot actions in local priorities and ensuring community-centered and sustainable urban transformation, insights from the workshop were directly transmitted to the UP2030 roadmap.

10.4. Public Awareness Activities

UP2030 project team hosted several public awareness activities, including the "Women's Ferry Event" focusing on vulnerable groups, and the "Climate Officers Event". The project was also introduced by dedicated stands and t-shirts. The team engaged with professionals and citizens, reaching approximately 1000+ people throughout these public awareness activities.

10.5. Upscaling Workshop

On June 26, 2025, Istanbul Metropolitan Municipality organized the UP2030 Upscaling Workshop at Müze Gazhane. The goal was to explore how the pilot applications developed in Kadıköy, Moda, such as digital twin modeling of building energy use and PV-integrated urban furniture, could be expanded to other districts in Istanbul. Participants were divided into four groups, each focusing on a different district: Beşiktaş, Ataşehir, Eyüpsultan, and Adalar. They discussed the local implementation of renewable energy systems, digital tools, and urban furniture. Topics included stakeholder roles, decision-making, implementation barriers, financing options, priority areas, and potential opportunities. A presentation on Kadıköy's energy model and a digital decision support tool helped guide the discussions, showing how energy renovation scenarios effect consumption, emissions, and cost. The workshop emphasized the importance of combining technology with local needs and governance, and prepared the basis for district-level action plans to support Istanbul's climate-neutral future.



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Stakeholders

EYODER: Enerji Verimliliği ve Yönetimi Derneği, Ege Üniversitesi, Tarım ve Orman Bakanlığı, BİMTAŞ, Şehir Plancıları Odası İstanbul Şubesi, MİMOTEK: Mikro Mobilite Teknolojileri Derneği, HOP, Eurosolar: Yenilenebilir Enerji Birliği, Parla Solar, TAT Gıda Sanayi Anonim Şirketi, ENVER: Enerji Verimliliği Derneği, Enerji ve Tabii Kaynaklar Bakanlığı, TENMAK, KİPTAŞ, Çevre Mühendisleri Odası, Bisikletliler Derneği, İSTON, GÜNDER: Uluslararası Güneş Enerjisi Topluluğu Türkiye Bölümü, TESCOM, İklim İçin 350 Derneği, Boğaziçi Üniversitesi, Çevre ve Şehircilik Bakanlığı, İstanbul İmar, TEİAŞ veya AYEDAŞ, Tüm Bisiklet Dernekleri Federasyonu, İBB Şehir Planlama Şube Müdürlüğü, GENSED: Güneş Enerjisi Sanayicileri ve Endüstrisi Derneği, Seven Sensor-Argesim Makine, TOFAŞ, Karbon Emisyonu Sera Gazı Düşürme ve Kontrol Derneği, ODTÜ, İlçe Belediyeleri, İstanbul Enerji A.Ş., Sürdürülebilir Mobilite İnisiyatifi Derneği, UGETAM, GÜNEŞDER: Güneş Enerjisi Yatırımcıları Derneği, Şişecam, TÜPRAŞ, İKLİMDER: İklim Enerji Çevre Sorunları Araştırma Derneği, İstanbul Teknik Üniversitesi, İstanbul Büyükşehir Belediyesi, Yaya Derneği, İSPARK, ELDER: Enerji Birlik ve Dernekleri, Smart Solar, Türk Traktör, Çevre İklim ve Enerji Eğitim ve Araştırmaları Derneği, Yıldız Teknik Üniversitesi, BUGEP: Bisikletli Ulaşımı Geliştirme Platformu, BELBİM, GÜYAD: Enerji Yatırımcıları Derneği, Solar3GW, Yorglass, Çevreci Enerji Derneği, İzmir Yüksek Teknoloji Enstitüsü, TEHAD: Elektrikli ve Hibrit Araçlar Derneği, İSBAK, ESDE: Enerji Stratejileri Derneği, ÇEDBİK: Çevre Dostu Yeşil Binalar Derneği, İBB Ulaşım Dairesi Başkanlığı, TENMAK, VERTE Elektronik, İGDAŞ, Zorlu Enerji, TÜBİTAK, İBB Enerji Yönetimi ve Aydınlatma Şube Müdürlüğü, TÜBİTAK MAM, İleri Ar-Ge Teknolojileri.

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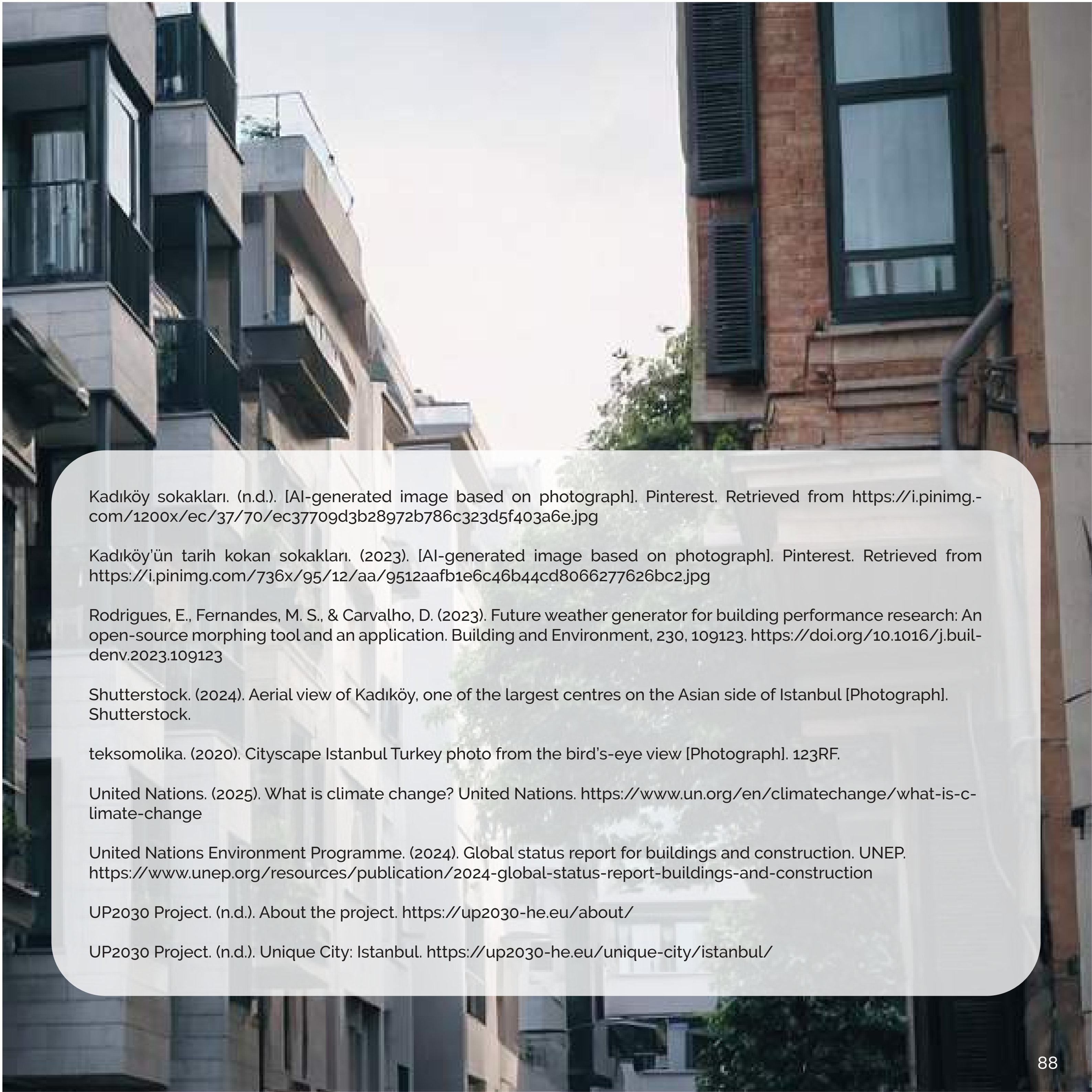

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